Artifact-Gated Evaluation of a Causal FIR Module for Sequential Recommendation

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We evaluate a small, detachable causal finite-impulse-response (FIR) residual for sequential recommendation on Amazon Reviews 2023 (5-core, full-catalog, leave-last-out). Matched-initialization internal estimates are positive on Musical_Instruments (+0.002265, 95% CI [0.001928, 0.002602]) and, under one untuned configuration, on Industrial_and_Scientific (+0.002110 [0.001820, 0.002399]) and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]). These are outcome-known or test-exposed filter-versus-identity studies, not independent confirmation. A pre-declared but likewise outcome-known six-arm control study sharpens the mechanism boundary: every temporally active arm improves identity after Holm adjustment, learned taps beat two algebraically redundant fixed-filter parameterizations, but learned-specific advantages over a shared-filter control (-0.000081 [-0.000337, 0.000175]) and a parameter-matched nonlinear causal control (+0.000204 [-0.000038, 0.000446]) are not established. Thus the evidence supports causal temporal mixing in this configuration, not uniquely learned per-channel FIR coefficients. Secondary findings are two environment-caveated per-category point-estimate threshold comparisons and a Musical_Instruments frequency-5-heavy text-tail effect; neither licenses a general system-ranking or cross-dataset claim. A fail-closed graph recomputes 190 paper-bound cells across 17 families and preserves null, deviated, and VOID outcomes alongside positive results.

CCS Concepts: • **Information systems → Recommender systems**; *Personalization.*

Additional Key Words and Phrases: sequential recommendation, Amazon Reviews 2023, HSTU, causal FIR filtering, long-tail recommendation, text-augmented recommendation, reproducibility, pre-declaration

1 Introduction

Sequential recommendation models predict a user's next item from their interaction history. SASRec [Kang and McAuley 2018] and BERT4Rec [Sun et al. 2019] established Transformer-based baselines. Later systems add language-derived item representations [Ding et al. 2021; Hou et al. 2024, 2022; Li et al. 2023] or discrete semantic IDs [Rajput et al. 2023; Yang et al. 2024].

Amazon Reviews 2023 results span incompatible preprocessing variants. In particular, the 0-core reference pipeline and the 5-core protocol used by recent generative retrievers do not define interchangeable benchmarks. Our experiments use the latter: fixed 5-core splits, leave-last-two-out histories, and full-catalog ranking.

Recommender evaluation is vulnerable to mistuned baselines, incompatible protocols, and irreproducible result selection [Ferrari Dacrema et al. 2021, 2019]. We therefore pair the method with a per-paper discipline: git-frozen protocols, a fail-closed graph that recomputes 190 reported cells across 17 claim families, environment-caveated comparator regeneration, and symmetric adjudication. Failures are retained. In particular, the original Office_Products protocol remains VOID after its own floor check exposed an environment mismatch, although a redesigned fresh-seed protocol later passed (§5.2; Appendix A.0).

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Using an HSTU-style pure-PyTorch encoder based on Zhai et al. [2024], we ask whether one small causal filtering component survives matched controls and transfers without category-specific tuning. Our contribution ledger is deliberately limited to three items:

(1) **A canonical causal FIR module.** We adapt earlier bidirectional frequency filters to a strictly left-causal, depthwise residual that is identity-initialized and gradient-active under the all-position objective (§3). This is a modular component, not a new recommender architecture.
(2) **Matched internal evidence and an active-control boundary.** Learned FIR–identity estimates are positive on Musical_Instruments (+0.002265, 95% CI [0.001928, 0.002602]) and under one untuned configuration on Industrial_and_Scientific (+0.002110 [0.001820, 0.002399]) and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]) (§5.2). Because these studies reuse outcome-visible splits, they are internal robustness evidence, not independent confirmation. On Musical_Instruments, shared and parameter-matched nonlinear causal controls also improve identity and are not separated from learned taps; this narrows the supported mechanism to causal temporal mixing rather than uniquely learned per-channel FIR coefficients. A frequency-5-heavy Musical_Instruments text-tail effect is secondary and does not establish cross-dataset heterogeneity (§5.3–§5.4).
(3) **An auditable evaluation record.** Released protocols, artifacts, sidecars where available, comparator-regeneration code, and a strict rebuild preserve positive, null, deviated, and VOID outcomes under the same reporting rule (§8). The apparatus certifies reconstruction and protocol history; it does not strengthen the estimand.

TAPE, late fusion, sparse-warm redistribution, titrations, probe screens, and earlier Beauty results are supporting studies. All borrowed architectures and training recipes are attributed in §2.

2 Related Work and Attribution

We summarize prior work that our experiments directly build on. A complete attribution table for every component used in our experiments appears at the end of this section.

SASRec [Kang and McAuley 2018] is the base Transformer sequential recommender we re-implement. SASRec models a user's history with a causal-masked Transformer encoder and scores the next item via dot product with a learned item-embedding table.

BERT4Rec [Sun et al. 2019] adapts BERT-style bidirectional masked-item modeling for sequential recommendation. We implement BERT4Rec as one of our 20 ablation baselines on Beauty_and_PC.

SBERT / MiniLM [Reimers and Gurevych 2019; Wang et al. 2020] provide pre-trained sentence embeddings. We use `sentence-transformers/all-MiniLM-L6-v2` [Sentence-Transformers maintainers 2021] (384-d) as the default frozen item-text encoder.

UniSRec [Hou et al. 2022] is an early, widely-adopted text-based sequential recommender that uses a pre-trained language model to embed item text and learns a parametric adaptor to project these embeddings into the sequential model's hidden space; **ZESRec** [Ding et al. 2021] precedes it in representing items from natural-language descriptions with a pre-trained encoder for zero-shot sequential recommendation, and **RecFormer** [Li et al. 2023] subsequently studies language representations for sequential recommendation including low-resource and cold-start evaluation — we make no priority claim in this line. The SASRec-with-frozen-text-encoder pattern we use is a strict simplification of UniSRec.

BLaIR [Hou et al. 2024] (published as Hou et al. 2026b) is a RoBERTa-base model continually pretrained on 10% of (item-metadata, review) pairs from Amazon Reviews 2023. We use `hyp1231/blair-roberta-base` (768-d) as an alternative frozen item-text encoder. Hou et al. also publish a reference implementation, **SASRecText**, that pairs SASRec with BLaIR via a 2-layer MLP adaptor `[768, 300, 64]` with Dropout(0.2) + ReLU; we adopt this adaptor design verbatim (see Appendix A.1).

TIGER [Rajput et al. 2023] reformulates sequential recommendation as autoregressive generation over discrete semantic IDs produced by a residual-quantized variational autoencoder over item-text embeddings. TIGER reports strong results on Amazon Reviews 5-core benchmarks.

LIGER [Yang et al. 2024] extends TIGER with a dense-retrieval refinement step and reports further gains.

Amazon Reviews 2023 [Hou et al. 2024] is the source dataset. We use the **5-core leave-last-out** protocol (described in §3) matching the convention used by TIGER, LIGER, and similar works, which differs from the `0core_timestamp_w_his_*` HuggingFace subset used by Hou et al.'s `seq_rec_results/` reference implementation.

Evaluation practice and reproducibility in recommender systems. Our lead contribution responds to a documented evaluation-trust problem in this field. Ferrari Dacrema et al. [2019] examined 18 recent neural recommenders and could reproduce only 7 with reasonable effort, 6 of which were often outperformed by comparably simple heuristic baselines; the extended journal analysis [Ferrari Dacrema et al. 2021] corroborates the pattern in greater depth and attributes it in part to the choice and optimization of the baselines used for comparison. Sun et al. [2020] respond at the community level with standardized benchmarking for reproducible evaluation and fair comparison. These works diagnose the problem and propose field-level remedies; what this paper adds is complementary and *per-paper*: an executable discipline that a single submission can adopt and a reviewer can re-run — version-controlled pre-declaration of the comparator-facing confirmation (§5.2), a fail-closed artifact gate that recomputes every artifact-gated table cell from released artifacts at build time (published comparator values are pinned as labeled external constants; §2.2 records a retirement the gate forced), local regeneration of the comparator by its own unmodified reference implementation (environment-caveated single-run regenerations, §5.6), and symmetric publication of failures, including a pre-declaration VOIDed by its own comparability check (Appendix A.0) with a redesigned successor that subsequently passed on fresh seeds (`PREREG_OFFICE_V3.md`; §5.2), the original VOID standing unchanged. None of this apparatus strengthens any empirical claim — every non-claim of §5–§7 stands unchanged; it certifies that the numbers printed are the numbers the artifacts recompute, under decision rules frozen before the confirmatory runs.

2.1 Attribution table

Component	Source	Where in our work
SASRec architecture	Kang and McAuley [2018]	base SASRecSBERT model
BERT4Rec	Sun et al. [2019]	one of 20 ablation baselines
MiniLM all-MiniLM-L6-v2 (384-d)	Reimers and Gurevych [2019]; Wang et al. [2020]	default frozen item-text encoder
BLaIR blair-roberta-base (768-d)	Hou et al. [2024] (arXiv:2403.03952)	alternative frozen item-text encoder
2-layer MLP adaptor [768, 300, 64] Dropout+ReLU	Hou et al. [2024] (SASRecText, in external/AmazonReviews2023/seq_rec_results/)	--mlp-adaptor flag
Rich-text content concatenation (title + categories + store name)	derived from Hou et al. [2024] preprocessing	encode_richtext_5core.py
Amazon Reviews 2023 dataset	Hou et al. [2024]	all benchmarks
TIGER / LIGER	Rajput et al. [2023] / Yang et al. [2024]	published comparator numbers

2.2 Our additions to the above prior art

We do not claim a wholly new recommender architecture. Our architectural additions are deliberately small: a causal FIR adaptation of prior frequency-filter ideas and a soft text-prototype additive term. Our contributions are:

- An **open-source 5-core preprocessing pipeline** (preprocess_5core_standard.py) that matches the protocol used by TIGER and LIGER (not the 0-core protocol used by Hou et al.'s repo).
- A **memory-efficient chunked-full-softmax loss** that enables proper full-catalog cross-entropy training at 207k-item catalogs on a single 16 GB GPU.
- An **eval optimization** (caching all_item_features() once per evaluate() call) that gives ~10× speedup at d=64 with the MLP adaptor.
- A **cross-pipeline transfer study**: empirically testing which published architectural choices help on the 5-core protocol.
- **Negative-result documentation** for 18 of 20 tested variants on Beauty_and_Personal_Care 5-core.
- The **competitive Video_Games 5-core result** on the HSTU-style pure-PyTorch stack (6-seed NDCG@10 = **0.0673** ± **0.0003**, +17.5% over published SASRec 0.0573), shown to be architectural (ID-only ≈0.0656), with no SOTA claim. *(The earlier v1-era SASRec-SBERT number 0.05509 ± 0.00035 has been retired from Table 1a: its raw per-seed artifacts predate the artifact manifest and cannot be recomputed, so it survives only as a RETIRED provenance row in the manifest, not as a paper claim.)*
- The two added components — the **strictly causal FIR temporal filter** (positive matched identity contrasts on three outcome-visible categories, with active controls narrowing attribution to causal temporal mixing; §5.2) and **TAPE** — plus the **MI frequency-5 tail case** (§5.3) and its **interaction-thinning titration** level-contrast result (§5.4). The novelty boundary for each is stated explicitly in Table 0 (§2.3).

2.3 Claim boundary

Table 0 separates prior work from the paper's actual claim boundary. Frequency filtering (including FEARec and WP-GRec) and causal convolution (including Caser and NextItNet) predate this work [Du et al. 2023; Liu et al. 2026; Tang and Wang 2018; Yuan et al. 2019], as do residual identity initialization, HSTU, frozen text encoders, and label smoothing. Our claim is limited to the gradient-active left-causal depthwise-FIR residual used before an HSTU-style stack under an all-position objective, together with its matched internal evaluation. The historical zero-gated realization is singular at initialization and remains package-attributed; only the nonsingular canonical realization supports a FIR-specific interpretation (§3, §5.2).

Table 0: Claim boundary — what is reused, what is changed, and what this paper claims.

Component	Prior art	What we reuse	What we change	Evidence	Claim boundary
HSTU base	Zhai et al. [2024]	pointwise `silu(QK`$^\top$ `+ rab) V` aggregation, per-block normalization, relative attention biases	pure-PyTorch implementation; diagnosis/removal of two spurious normalizations in our initial reimplementation; equation-level verification only, core-block parity demonstrated at a mirrored aligned configuration (max abs diff 0.0; §3.2 scope note — the trained block is a strict-superset variant); no pinned end-to-end system reproduction (§5.6, §6.3)	Table 1 ablations; §3.7	prior-work implementation
BLaIR / SBERT text embeddings	Hou et al. [2024]; Reimers and Gurevych [2019]; Wang et al. [2020]	frozen encoders; Hou et al.'s MLP adaptor verbatim	nothing (frozen, off-the-shelf)	Table 1a encoder comparison	prior work
Label smoothing	Szegedy et al. [2016]	standard uniform label smoothing	applied to full-catalog chunked-softmax CE in this setting	+0.0013 single-flag; +0.0012 stacked (Table 1)	prior work
Time bias	TiSASRec [Li et al. 2020]; HSTU [Zhai et al. 2024]	bucketed time-interval attention bias	integration only	+0.0027 single-flag, largest classical component (Table 1)	prior work
FMLP-Rec / BSARec frequency filters	Zhou et al. [2022]; Shin et al. [2024]	the idea of filtering sequential representations; frequency-domain motivation; attention-oversmoothing framing	not inserted as-is (bidirectional in their original formulations)	cited motivation (§3.7b)	prior work

Component	Prior art	What we reuse	What we change	Evidence	Claim boundary
Causal FIR adaptation (ours)	builds directly on FMLP-Rec / BSARec; causal *convolutional* sequence encoders are prior art (Caser; NextItNet)	their filtering motivation and frequency framing	minimal leak-free, left-causal depthwise FIR residual, Δ=0 identity-initialized yet gradient-active, inserted before an HSTU-style stack under the all-position next-item objective, full-catalog LLOO	outcome-known matched-init FIR−identity estimates: MI +0.002265 [0.001928, 0.002602], IS +0.002110 [0.001820, 0.002399], CDs +0.006150 [0.005849, 0.006450]; active-control study supports temporal mixing but not uniquely learned taps (§5.2)	incremental modular contribution
TAPE (ours)	TIGER [Rajput et al. 2023]; VQ-Rec; ProtoMF	prototype / semantic-ID shared-structure motivation; frozen text embeddings	frozen k-means soft assignment gating a zero-init learnable prototype table, additive	+0.0009 single-flag (n=1); +0.0004 in the 4-seed cross-check; sub-additive (§5.1)	supporting ablation
Tail / thinning analysis (ours)	popularity-stratified evaluation (standard); long-tail SR, e.g. MELT [Kim et al. 2023, arXiv: 2304.08382]; cold-start content methods, e.g. DropoutNet [Volkovs et al. 2017], CLCRec [Wei et al. 2021]	nominal train-frequency tercile bucketing (§5.3 cohort caveats); same-seed-number contrasts (not initialization-paired)	per-dataset (heterogeneity not established) text-vs-ID tail contrast plus interaction- and user-mode thinning interventions with a matched-R1 (resource-controlled) design	MI tail win replicated by the pre-declared, outcome-visible repaired-estimand TFV2 campaign (+0.000420, Holm-PASS; not confirmatory — §5.3); VG null (equivalence claims retracted §5.3); cross-dataset contrast not significant under TFV2 (p = 0.13, descriptive); user-mode dd = +0.000326 (suggestive, p = 0.058, §5.4.2)	secondary empirical boundary

Closest systems I — adjacent lines and the narrowed boundary (metadata verified 2026-07-19). Five adjacent lines narrow what this paper may claim, and we state the narrowed boundary explicitly. (i) *Text similarity for rare/unseen items:* SimRec [Brody and Lagziel 2024] already targets sequential-recommendation cold-start with text-derived item similarity and reports larger benefits on sparser datasets; our tail contribution is therefore graded **incremental** — a full-catalog, cross-category AR2023 extension with intervention studies and disclosed cohort defects (§5.3–§5.4) — not a new sparse-vs-dense semantic-tail idea. (ii) *Convolution/filtering with attention:* C3SASR [Chen et al. 2022] combines cheap causal convolutions with self-attention, AdaMCT [Jiang et al. 2023] mixes local convolution with Transformer attention, and learnable frequency-domain filtering with gates and side-information fusion appears in DLFS-Rec [Liu et al. 2023a], DIFF [Kim et al. 2025], TASIF [Luo et al. 2026], and — with position-specific time-variant convolutional filters — TV-Rec [Shin et al. 2025]; our defensible distinction is only the **gradient-active, identity-initialized left-causal depthwise FIR residual under an HSTU-style all-position objective at full catalog** — not causal/local convolution with attention, learnable filtering, gating, or filter-content fusion as concepts. (iii) *Side-information fusion scope:* NOVA [Liu et al. 2021] and DIF-SR [Xie et al. 2022] document that direct/early fusion can be harmful and propose decoupled designs; TedRec [Xu et al. 2024] fuses text and ID at sequence level in the frequency domain; and AlterRec [Li et al. 2025] studies failures of naive ID/text fusion — we test **only early additive frozen-text**

fusion, and every "text helps/does not help" statement in this paper is scoped to that construction. (iv) *Cold-start distinction (stated precisely):* Mecos [Zheng et al. 2021] meta-learns for **new items with few interactions**, and RecGPT [Jiang et al. 2025] is a generative foundation model evaluated in **zero-shot transfer and short-history** settings — two different cold-start settings, both distinct from our mixed zero-/low-exposure full-catalog cohort (which itself mixes zero-train-exposure and low-frequency targets, §5.3); we make no cold-start-method comparison. (v) *Reproducibility apparatus:* executable evaluation frameworks (Elliot [Anelli et al. 2021]; DaisyRec 2.0 [Sun et al. 2022]), accountability workflows [Bellogín and Said 2021], and reproduction audits [Petrov and Macdonald 2022] predate this work; what this paper claims is only its exact combined enforcement mechanism — per-cell recomputation from released artifacts, provenance tripwires, and symmetric self-VOIDing adjudication — positioned against, not above, those frameworks. TAPE's boundary likewise narrows: LLMSQRec [Zhang et al. 2026] combines frozen LLM semantic embeddings, ID embeddings, and learned prototype/codebook quantization for sparse and long-tail settings — TAPE's continuous additive realization differs, but the broad text-prototype-ID concept is not ours. Relatedly, the chunked-full-softmax loss is implementation engineering; memory-efficient exact cross-entropy without materializing the full logit matrix is published (Cut Cross-Entropy [Wijmans et al. 2025]), and we cite it as such (§3.5). **Closest systems II — null-space, frequency, and anisotropy lines (metadata verified 2026-07-20).** Five further works tighten the frozen-text/ID and frequency boundaries: **AlphaFuse** [Hu et al. 2025] learns ID embeddings in the null space of language embeddings and evaluates long-tail settings — it is now the **closest omitted frozen-text-plus-ID comparator**, and benchmarking it under our full-catalog split (or an explicit protocol-based exclusion) is queued experiment work; **DWSRec** [Zhang et al. 2024] whitens anisotropic pre-trained text embeddings (dual-view); **SIDSRec** [Zhou et al. 2026] separates semantic and ID channels before fusion; **BFDRec** [Hu et al. 2026] does energy-aware multi-scale frequency modeling with dynamic gating (closest frequency-domain line beside TASIF/DLFS-Rec/DIFF/TV-Rec); and **ACE** [Lee et al. 2026] shows a linear-autoencoder reshaping of anisotropic LLM embeddings can itself add capacity usefully — so our negative map's capacity-adding conclusions remain strictly **implementation- and environment-local** (they characterize our probes in our setting, not the design space). Two protocol **controls remain open experiments (disclosed, not executed):** a sequence-splitting/target-strategy parity audit mapping our all-position objective onto the taxonomy of the SIGIR 2026 reproducibility study (arXiv:2604.05309) with demonstrated comparator target-multiplicity parity, and an item-text permutation control (attribute-shuffling style, DOI 10.1145/3805712.3809839) before attributing the +2.7% stack lift to semantic alignment rather than added representation structure.

Closest systems III — LLM-semantic tail and convolutional lines (metadata verified 2026-07-20). Four further works narrow the two axes our findings sit on. *Semantic-tail line:* **LLM-ESR** [Liu et al. 2024] (Liu, Wu, Wang, Zhang, Tian, Zheng & Zhao — NeurIPS 2024 spotlight) fuses LLM semantic embeddings with collaborative signals in a dual-view design for long-tail items and adds retrieval-augmented self-distillation for long-tail users; **FAERec** [Wei et al. 2026] (Wei, Dang, Guo, Zhao & Sun, 2026) adaptively gates ID and LLM embeddings with dual-level (contrastive + correlation) alignment for tail items; **LLM2Rec** [He et al. 2025] (He, Liu, Zhang, Ma & Chua — KDD 2025) trains a two-stage LLM-derived embedding model encoding both semantics and collaborative signal. LLM-ESR and FAERec target long-tail improvement directly, and LLM2Rec targets general sequential recommendation via LLM-derived embeddings — an active methods axis. **LLM2Rec is the closest protocol overlap**: it evaluates on Amazon Reviews 2023 with 5-core filtering, leave-one-out and full-item ranking, but with maximum history length 10 (ours: 50) and a correspondingly different item universe and splits, so its numbers are not comparable under the §2.1 executable-protocol boundary; LLM-ESR, FAERec, and ConvFormer do not use this protocol. None of this changes what we claim: our tail contribution is not a method but a *measured boundary* — the MI frequency-5-heavy pattern (cross-dataset heterogeneity not established) of a deliberately frozen additive text stack against an ID-only control under a gated protocol (§5.3), including where text does *not* help (zero-exposure targets: zero hits through rank 100 in all four TFV2 categories). Whether these adaptive-fusion designs move the same frozen-protocol boundary is an open experiment we have not run, and we say so. *Convolutional-sequence line:* **ConvFormer** [Wang et al. 2023] (Wang, Lian, Wu, Li, Fan, Xu, Li & Xie, 2023) replaces item-to-item attention with convolution in sequential user modeling — the nearest Transformer-replacing convolutional relative alongside FMLP-Rec, BSARec, AdaMCT, and C3SASR; our FIR claim remains exactly the narrow one already stated (a gradient-active, identity-initialized leak-free left-causal FIR residual *before* an HSTU-style stack under the all-position objective, supported by matched-initialization contrasts on three categories), not a claim to convolutional sequence modeling as such.

3 Method

3.1 Preprocessing pipeline

We preprocess Amazon Reviews 2023 [Hou et al. 2024] into 5-core leave-last-out splits. For each category we:

1. Load raw user-item review interactions from the `raw_review_<category>` HuggingFace subset.
2. Apply a 5-core filter iteratively until no user or item has fewer than 5 interactions.
3. Sort each user's interactions chronologically.
4. Hold out the last interaction as test, the second-to-last as validation, the rest as training.
5. Encode item metadata (`meta_<category>.jsonl`) with the chosen frozen text encoder.

This protocol matches TIGER [Rajput et al. 2023] and LIGER [Yang et al. 2024] but differs from Hou et al. [2024]'s reference repo (`external/AmazonReviews2023/seq_rec_results/`) which uses the `0core_timestamp_w_his_*` HuggingFace subset. The 5-core protocol produces smaller, denser interaction matrices.

Estimand and construct boundary. Each review event is treated as an implicit positive interaction regardless of its 1–5-star rating or verified-purchase flag. The estimand is therefore *ranking the next recorded review event among the fixed post-filter catalog*, not predicting preference, satisfaction, purchase, or deployment engagement. Five-core eligibility, item indices, and the candidate catalog are computed from complete histories before splitting, so future events can affect inclusion even though their identities are not used as training targets. Rating-threshold, verified-purchase, genuine implicit-event, and global-time/query-time-catalog sensitivities have not been run; conclusions do not extend to those constructs or deployment regimes.

Pre-core deduplication (disclosed 2026-07-19). Before k-core filtering, duplicate `(user, item)` interactions are collapsed to a single event, keeping the **earliest** timestamped occurrence (first-exposure convention). This step is material and was previously undocumented in the paper: it removes 41,888 of 3,017,439 raw rows on Musical_Instruments (1.39%), 69,115 of 4,624,615 on Video_Games (1.49%), 320,078 of 23,911,390 on Beauty_and_PC (1.34%), and 156,363 of 12,845,712 on Office_Products (1.22%); the FIR-breadth categories were preprocessed from AR2023 rating-only exports whose (user, item) pairs are already unique (0 rows removed on Industrial_and_Scientific and CDs_and_Vinyl). Deduplication can change k-core membership, sequence order, and held-out targets; sensitivity to a keep-latest rule is untested (a disclosed limitation). (The MI/Office counts are from the retained preprocessing logs; the VG/Beauty logs were not retained, and their counts were regenerated 2026-07-19 from the raw dumps with the released deterministic script.)

Timestamp-tie policies (disclosed 2026-07-19; two distinct deterministic rules). Two tie-breaking rules operate at different stages and were previously undocumented. (i) Preprocessing stable-sorts each user's interactions by timestamp only, so for equal-timestamp events the raw-file order decides both which duplicate survives deduplication and which event lands on a leave-two-out split boundary. (ii) The trainer later re-sorts each user's train sequence by `(timestamp, item_index)`, so within-train equal-timestamp order is decided by reindexed item ID. Both rules are deterministic given the released raw dumps and code. Measured scale (2026-07-19, from the released splits): on Musical_Instruments, 3 of 57,439 users (0.01%) have test-target timestamp equal to the validation-target timestamp, 3 have validation equal to the last train timestamp, and 12 (user, timestamp) groups cover 24 of 511,836 rows; on Video_Games the counts are 4 and 8 of 94,762 users and 61 groups covering 122 of 814,586 rows (0.01%). At this scale the tie policies cannot materially affect any reported number; we nevertheless document them as part of the exact preprocessing contract.

3.2 Model architecture

Shared item-feature construction (both encoder families). For each item `i`:

- A learned per-item embedding $e_i \in \mathbb{R}^d$ (initialized N(0, 0.02), padding-aware).
- A frozen text embedding $s_i \in \mathbb{R}^k$ from MiniLM (k=384) or BLaIR (k=768).
- A learnable projection $\Phi : \mathbb{R}^k \to \mathbb{R}^d$.

The item feature passed into the encoder is $f_i = e_i + \Phi(s_i)$. Sequences are right-padded with a dedicated pad token and processed with a causal-only attention mask (no key-padding mask — see §3.4). The output logits for position t are $h_t \cdot$ `all_items.T` where `all_items` is the (n_items, d) item-features table.

Headline encoder — HSTU-style (every §5 result). All headline experiments use the **HSTU-style pure-PyTorch pointwise-attention encoder**, described in full in §3.7: pointwise silu$(QK^\top + \text{rab})V$ aggregation with the published per-block normalization — no softmax, no $1/\sqrt{d}$ scaling — with the core block verified exactly (max abs diff 0.0) against the reference research implementation at a mirrored, aligned, dropout-free configuration (`HSTU_PARITY_REPORT.md` discloses the alignment; the trained configuration is a strict-superset HSTU-style variant — §6.3). Configuration, stated here and in §4.4: **4 layers, 2 heads, d = 64, dropout 0.5**.

Baseline encoder — SASRec-SBERT (protocol parity and the superseded Appendix A.1 scans only). The SASRec-family baseline follows Kang and McAuley [2018]: a 2-layer pre-norm Transformer encoder (d=64, n_heads=2, FFN dim=4d, GELU activation) followed by a final LayerNorm, using the same item-feature construction. **No headline number uses this encoder**; it appears as the per-category parity/floor baseline and in the superseded Beauty scan (Appendix A.1).

3.3 Choice of projection Φ

Following our cross-pipeline transfer study (Appendix A.1), we compare two designs for Φ:

- **Linear**: $\Phi(s) = Ws$ for a learned weight $W \in \mathbb{R}^{d\times k}$.
- **MLP** (adopted from Hou et al. [2024] SASRecText): $\Phi(s) = W_2 \cdot \text{ReLU}(W_1 \cdot \text{Dropout}(s))$ with hidden dim 300 and Dropout(0.2) between and before the linear layers. **This design is from Hou et al. [2024] and we do not claim novelty for it.**

3.4 Right-padding + causal-only mask

During exploration we identified a numerical failure mode in the standard SASRec implementation pattern: combining left-padding with both a key-padding mask and a pre-norm Transformer encoder produces NaN logits when an entire row of keys is padded (the softmax denominator is zero). The NaN propagates through residual connections and silently inflates eval NDCG to ~0.9955 — a "fake-perfect" result. We adopt right-padding + causal-only mask (no key-padding mask) to avoid this trap.

3.5 Chunked-full-softmax loss

For training with cross-entropy over the full catalog (no sampled negatives), the full logits tensor `(B, L, n_items)` is prohibitive at L=50 and n_items=207k. We implement an incremental-logsumexp loss:

```
def chunked_full_softmax_loss(hidden, targets, model, item_chunk):
    lse = -inf  # running logsumexp over the item dimension
    target_scores = None
    for chunk_start in range(0, n_items, item_chunk):
        chunk_emb = model.item_features(chunk_ids)
        chunk_logits = hidden @ chunk_emb.T
        lse = logsumexp(lse, logsumexp(chunk_logits, axis=-1))
        if any(target in chunk):
            target_scores[in_chunk] = chunk_logits[in_chunk, target]
    return -mean(target_scores - lse)
```

This bounds peak memory by `(batch, item_chunk)` instead of `(batch, n_items)` and allows proper full-softmax training at the 207k-item Beauty_and_Personal_Care scale on a 16 GB GPU.

Memory-efficient exact cross-entropy has published precedent (Cut Cross-Entropy, ICLR 2025, computes exact CE without materializing the full logit matrix); our chunked variant is independent implementation engineering in the same spirit, claimed only as engineering.

3.6 Evaluation protocol

We score each user against the **full catalog** of 207k items (Beauty_and_PC) or 25k items (Video_Games). For the held-out test item, we compute the rank under dot-product scoring (after masking train+val items the user has already seen) and report NDCG@10, HR@10, MRR. We use the standard SASRec test-time convention of including the validation item in the input sequence [Kang and McAuley 2018].

For our default 16 GB GPU, the optimized evaluation caches the (n_items, d) item-features table once per `evaluate()` call rather than recomputing it per batch — this is critical with the MLP adaptor, which would otherwise recompute the 207k-item $\times$ 768 $\rightarrow$ 300 $\rightarrow$ 64 MLP forward 1,400+ times per evaluation.

3.7 Added components (deliberately small adaptations)

Our two architectural additions (the novelty boundary is stated in §2.3) are deliberately simple, default-OFF, and zero-initialized so that each is a *bit-identical no-op at initialization* and reduces the design to a single controlled lever. Both are defined on top of an **HSTU-style pure-PyTorch implementation of the HSTU encoder, based on the published**

architecture [Zhai et al. 2024]: we use the pointwise-aggregated interaction $\mathrm{silu}(QK^\top + \mathrm{rab})\,V$ with the published per-block normalization, having diagnosed and removed two spurious normalizations (a $1/\sqrt{d}$ score scaling and a $1/(i+1)$ row averaging) in our initial reimplementation that collapse HSTU's pointwise attention into weak mean-pooling. We verify the implementation equation-by-equation against the published HSTU formulation (pointwise $\mathrm{silu}(QK^\top + \mathrm{rab})\,V$ aggregation, no softmax, no $1/\sqrt{d}$ scaling); an executable end-to-end reproduction of the official kernel stack is impossible on our hardware (§5.6); the core block, however, passes an exact numerical parity test against the reference research implementation at a mirrored **aligned configuration only** (identity affine norms, zeroed extra `uvqk` bias, ε aligned, dropout off, eval mode; max abs diff 0.0; `HSTU_PARITY_REPORT.md` discloses the alignment — the trained configuration is a strict-superset HSTU-style variant, §6.3). **Initialization disclosure (2026-07-19):** with the delta-kernel and gate g=0 initialization, both task gradients are exactly zero at the start (y=x, so the gate gradient vanishes and the kernel gradient is multiplied by g=0); learning bootstraps only because Adam's coupled weight-decay term (1e-5) perturbs the delta tap off its fixed point — the optimizer regularizer is therefore part of the filter's learning mechanism, and with weight decay disabled this parameterization is an absorbing no-op. The fixed-average and no-gate comparator arms do not share this singular start, so those comparisons alter the optimization path as well as the named design element. The later nonsingular studies provide cleaner internal contrasts but reuse outcome-visible splits, and active controls prevent uniquely learned-tap attribution; they do not retroactively decompose the non-initialization-matched historical arms.

(a) Text-Anchored Prototype Embeddings (TAPE). TAPE is a simple soft text-prototype additive capacity term inspired by semantic-ID and prototype methods (TIGER, VQ-Rec, ProtoMF); it provides a small but not headline gain (§5.1). Let $t_i \in \mathbb{R}^{d_{\text{text}}}$ be item i's frozen text embedding (MiniLM or BLaIR). We k-means the L2-normalized text embeddings into K centroids $\{c_1, \dots, c_K\}$ once, offline, and form a **frozen soft assignment** $A_i = \mathrm{softmax}(\langle t_i, c\rangle/\tau) \in \Delta^K$ (τ = 0.05). We add a **learnable prototype table** $P \in \mathbb{R}^{K\times d}$ (zero-initialized) to each item's feature:

$$e_i = \mathrm{id}_i + \Phi(t_i) + A_i P,$$

where id_i is the learned per-item embedding and Φ the learned projection of the frozen text embeddings. Items in the same semantic neighborhood share the learnable capacity $A_i P$ — the shared-structure benefit of TIGER-style semantic IDs [Rajput et al. 2023] without discrete codes or an autoregressive decoder. Because A is frozen and P is zero-init, TAPE is an exact no-op at initialization and recovers the baseline. TAPE is adjacent to TIGER (discrete RQ-VAE IDs), VQ-Rec [Hou et al. 2023] (PQ codes), and ProtoMF [Melchiorre et al. 2022] (prototype matrix factorization for explainability); it differs only in realization — a frozen text-derived soft assignment gating a learnable prototype table inside a sequential recommender — and we present it as a small adaptation of these prior ideas tested as a secondary ablation, not as a headline contribution. Default K = 512.

(b) Canonical causal FIR temporal filter (the proposed module). BSARec [Shin et al. 2024, Thm 3.1] analyzes vanilla Transformer self-attention as low-pass in the sequential-recommendation setting. We hypothesize that the HSTU-style pointwise aggregation used here can benefit from a similar local temporal inductive bias; the empirical ablation (§5.1–§5.2) tests this hypothesis. The proposed module is a single **per-channel learnable FIR residual** on the sequence embeddings $x \in \mathbb{R}^{B\times L\times d}$, in a **nonsingular, gradient-active** parameterization:

$$x \leftarrow x + \mathrm{DWConv}_\Delta(\mathrm{pad\_left}(x, K{-}1)), \qquad \Delta = 0 \text{ at init},$$

implemented as a depthwise `Conv1d(d, d, kernel_size=K, groups=d, bias=False)` whose tap residual Δ is zero at initialization, with a **left-only causal pad of** $K{-}1$, so the layer is an **exact identity at initialization** yet has

nonzero gradients on the first task update — there is no gate whose zero initialization creates an absorbing fixed point. The primary E-A and canonical-breadth learned arms used the backbone weight-decay setting for FIR taps; E-A's zero-FIR-weight-decay sensitivity was not detectably different (A2−A1 +0.000010 [-0.000339, +0.000360], $p = .95$), which is neither an equivalence result nor a pathway/mediation test (§5.2). The left-only pad makes output position t a function of input positions $\leq t$ only — so under our all-position next-item loss there is **no future-label leakage**. FMLP-Rec [Zhou et al. 2022] and BSARec use bidirectional sequence filters in their original formulations; under our all-position next-item loss, directly inserting such filters before each position's prediction would mix future positions, so we use this left-causal realization. **Legacy zero-gated realization (historical runs; reported as package-level).** The Video_Games headline configuration (§5.1) and the original breadth campaign (§5.2) were trained with a functionally related but *singular* realization — $x \leftarrow x + g \odot (y - x)$ with a delta kernel and a scalar gate $g = 0$ at initialization — in which both task gradients vanish at the start, so learning could begin only after Adam's weight-decay term perturbed the delta tap (§3 initialization disclosure). We call that the **legacy zero-gated FIR package** and report its contrasts as package-level (FIR-plus-initialization/optimizer), not FIR-specific. The nonsingular canonical form was then evaluated under matched per-seed initialization on three already outcome-visible category/split settings. Those internal estimates are positive, but their test exposure prevents an independent-confirmation claim; the active-control study further narrows the mechanism to causal temporal mixing rather than uniquely learned per-channel taps (§5.2). The novelty boundary is narrow and we state it plainly. *Not new*: filtering sequential representations, causal convolution (Caser; NextItNet), the frequency-domain motivation, and the attention-oversmoothing framing (FMLP-Rec; BSARec). *New*: the minimal, gradient-active, identity-initialized, leak-free left-causal depthwise-FIR residual inserted before an HSTU-style stack under the all-position next-item objective, evaluated under full-catalog LLOO. As an FIR filter the module has a learnable per-channel frequency response — the only sense in which we use the word "spectral" — and it supplies an explicit local temporal bias that the baseline does not learn as reliably in our low-capacity setting (we do not claim the attention stack cannot represent this behavior). The legacy-package gain is robust across kernel lengths K ∈ {4, 8, 16, 50} (§5); the canonical primary configuration fixes **K = 16** (historical runs used K = 8 on Video_Games and K = 16 on Musical_Instruments, §5.1–§5.2).

Both components are leak-free by construction and add negligible parameters (TAPE: K×d ≈ 33k; filter: K×d ≈ 0.5–3k). Two further regularizers we use are *not* claimed novel: **label smoothing** [Szegedy et al. 2016] on the full-softmax CE target, and the standard time / text-similarity / relative-position attention biases (TiSASRec; HSTU; Shaw et al. 2018), each cited at the point of use.

4 Experiments

4.1 Datasets

All datasets are AR2023 5-core categories under the same full-catalog LLOO protocol (§3.6). Counts are **total 5-core interactions** (train = total − 2×users, since LLOO holds out one valid and one test interaction per user):

Category	Role in this paper	Users	Items	Interactions (total)
Video_Games	headline reference numbers (§5.1); tail-pattern null (equivalence claim retracted §5.3)	94,762	25,612	814,586
Musical_Instruments	cross-category FIR confirmation + **pre-declared comparator confirmation** (§5.2)	57,439	24,587	511,836
Office_Products	V1 prereg **VOID** (descriptive; Appendix A.0); **redesigned V3 prereg PASSED** (§5.2; `OFFICE_V3_RESULTS.md`)	223,308	77,551	1,800,878
Beauty_and_ Personal_Care	tail-pattern exploratory 3-seed estimate (§5.3); superseded cross-pipeline scan (Supplement S.1)	729,576	207,649	6,624,441
Industrial_and_Scientific	**pre-declared FIR-breadth: artifact-PASS** (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; `FIR_BREADTH_RESULTS.md`)	50,985	25,848	412,947
CDs_and_Vinyl	**pre-declared FIR-breadth: artifact-PASS** (frozen rule fired; paired premise withdrawn; post-hoc independent-arm estimate — §5.2; `FIR_BREADTH_RESULTS.md`)	123,876	89,370	1,552,764

Video_Games/Musical_Instruments/Office_Products match the HSTU-BLaIR comparator pipeline's own statistics exactly on users/items (interactions within ±1; §5.2). The Beauty_and_PC catalog is 8× larger than Video_Games, with median user history of only 5 interactions — a harder benchmark. Every pre-declared campaign that has completed is mechanically adjudicated (verdicts at the relevant result blocks: §5.2, §5.7, §5.8, and Appendix A.0).

4.2 Baselines

- Popularity (trivial floor)
- SASRec with sampled-512 softmax (our "original" baseline)
- SASRec with chunked-full-softmax (improved baseline)
- BERT4Rec [Sun et al. 2019]
- Published comparators: TIGER [Rajput et al. 2023], BLaIR [Hou et al. 2024], LIGER [Yang et al. 2024]

4.3 Experimental design (headline runs)

All headline results (§5.1–§5.4) use the HSTU-style pure-PyTorch encoder (§3.2) under the fixed AR2023 5-core full-catalog LLOO protocol (§3.6). Each configuration is trained for **40 epochs** (the §5.1 Video_Games headline; the pre-declared-file MI/Office V3/FIR-breadth campaigns train 20 epochs per their frozen configurations) at batch size 256 (d_model 64, 4 layers, 2 heads, dropout 0.5) with the memory-efficient chunked-full-softmax loss (§3.5, item chunk 32,768) and a warmup-cosine learning-rate schedule. On top of the bias stack (TAPE-512 + time bias + text-similarity bias + pos-rab) the winning configuration adds label smoothing (ε=0.2) and the causal FIR filter (K=8). We run **6 seeds (20260608–20260613)** for the full-model headline and **5 seeds (20260608–20260612)** for every per-component ablation rung, reporting mean ± sample-std; the reported test number is always selected by best validation NDCG@10 (best-by-val), never by test. **Fixed-split inference scope (stated prominently, 2026-07-20):** the training seed is the only randomized unit; every interval in this paper quantifies optimization variability on one fixed public split per category — nothing here estimates user-resampling, split-choice, category-sampling, or cross-dataset uncertainty, and all claims are scoped to these exact datasets and splits. Evaluation is full-catalog (n_eval = 94,762 on Video_Games), with all train+val items masked (§3.6).

The tail analysis (§5.3–§5.4) rests on two arm comparisons, each toggling one named configuration flag on the same split and seed numbers (bundled interventions: the arms are not initialization-paired and optimizer paths differ) and reporting `by_popularity` NDCG@10 / HR@10 on leak-free train-frequency terciles (terciles frozen at full density):

- **text vs ID-only:** the full text stack against an ID-only ablation (`--no-sbert`, no prototypes, no text-similarity bias).
- **density / connectivity titration (§5.4):** training-only sub-sampling that thins either interactions (interaction-mode) or whole users (user-mode) to match a sparser reference category's resource levels, with the evaluation set held fixed.

A separate **20-variant cross-pipeline transfer scan on Beauty_and_PC** — supporting reproducibility material, not part of the headline spine — is reported in Appendix A.1.

4.4 Training details and hardware

Headline runs use Adam (lr 1e-3, weight decay 1e-5) with gradient clipping (norm 5.0) under the 40-epoch warmup-cosine schedule above. Hardware: a single NVIDIA RTX 5060 Ti (16 GB, Blackwell sm_120). Video_Games training takes ~10 min/seed; Beauty_and_PC ~1.5–2 hr/seed with the chunked-full-softmax loss. (The Appendix A.1 Beauty scan used a shorter 15–20-epoch budget and seeds 42/43; those details are local to that supporting study.)

5 Results

Statistical reporting conventions. Unless stated otherwise: the sample unit is the training seed; means are reported ± sample standard deviation over seeds; confidence intervals are two-sided 95% Student-t intervals with df = n−1 (the six-seed Video_Games headline is n = 6; controlled ablation rungs are generally n = 5; every other exception is named at the point of use); comparisons between our own arms use same-numbered seeds and are analyzed as independent arms (Welch/Satterthwaite) — same-seed-number arms are NOT initialization-paired (§5.3 randomization disclosure; per-seed difference tables are descriptive); comparisons against published numbers are point-estimate comparisons (the comparator is single-seed and unpaired); pooled tail hit-counts are reported as descriptive counts only — the same tail users recur under every seed and under both arms, so two-proportion z statistics over seed-summed pseudo-trials are invalid and are retracted (2026-07-19); the accompanying inference treats the five trained models per arm as the units (independent-arm Welch, §5.2/Appendix A.0); p-values are uncorrected unless a correction is named — cells labeled *confirmatory* in the artifact manifest are pre-declared prospective campaigns or their frozen-protocol locks — selection timing, not seed count, is the criterion (corrected 2026-07-19; the graph-taxonomy audit against this criterion is queued) — while multi-seed post-hoc results improve precision without becoming confirmatory, and *exploratory* cells carry no confirmatory weight.

5.1 Video_Games — multi-seed reference numbers (NOT SOTA)

Our headline result (Table 1) is built on the **HSTU-style pure-PyTorch encoder** with the two added components and stacks each lever in a controlled 5-seed ablation; Table 1a reports the SASRec-family text baselines that establish protocol parity; Table 1b compares to the reported HSTU-BLaIR reference and our SM120 compatibility-port reproduction.

Table 1: Headline component ablation — NDCG@10 on AR2023 Video_Games 5-core LLOO (full-catalog eval, n_eval = 94,762; 5 seeds = 20260608…20260612 unless noted; the full-model row is 6-seed 20260608…20260613; one flag added per row).

Configuration	NDCG@10	seeds	Δ
HSTU-style encoder, plain	0.0588	1	—
+ TAPE-512 (sub-additive text component)	0.0597	1	+0.0009
+ full bias stack (TAPE + time + text-sim + pos-rab)	0.0637 ± 0.0003	5	+0.0049 vs plain (+8.3%)
+ label smoothing ε=0.2 [Szegedy et al. 2016]	0.0649 ± 0.0003	5	+0.0012 (bands non-overlapping)
+ causal FIR filter K=8 (ours) → full model	**0.0673 ± 0.0003**	**6**	**+0.0024 (bands non-overlapping)**
(single-arm) FIR package arm (filter component; attribution open), no label smoothing	0.0652 ± 0.0003	5	+0.0015 vs the 0.0637 stack
(single-arm) ID-only (no SBERT / no text-sim / no prototypes)	≈0.0656 ± 0.0002	5	text adds only +0.0018 (+2.7%) overall

The full model is **+17.5%** over the published SASRec baseline (0.0573, Table 1b) on the identical protocol — but **this win is architectural, not text-driven**: the ID-only ablation already reaches ≈0.0656 (itself +14% over published SASRec), and the entire frozen-text stack (SBERT features + text-sim bias + prototypes) adds only **+0.00178** ± **0.00021 (5-seed, +2.7%)** overall. The two supported regularizers stack with a combined lift whose excess over the component sum is 33% — additivity and orthogonality were NOT tested (loss target vs. embedding spectrum): label smoothing alone +0.0012, causal filter alone +0.0015, combined +0.0036 over the bias stack. Per-component single-flag attribution on the HSTU-style implementation: the **time bias is the largest classical component (+0.0027)**, label smoothing +0.0013, causal filter +0.0015, while **text-similarity bias is dead weight (±0.0001, drop candidate)** and **TAPE is sub-additive (+0.0009)**. *(The time-bias, text-sim, and TAPE single-flag figures here are the single-seed seed-20260608 DECOMP values, honestly flagged n=1 (reported inline here, not in a table). A 4-seed multi-seed cross-check — DECOMP5, seeds 20260609–12, HSTU-style base 0.0594 — confirms the ordering: time bias +0.0030 (largest classical component, $\geq$ the single-seed +0.0027), pos-rab +0.0012, TAPE +0.0004, text-sim −0.00005 (no benefit observed at tested power). The qualitative attribution is unchanged; the only number that materially moves is TAPE, whose multi-seed single-flag lift (+0.0004) is smaller still than the single-seed +0.0009 — further support for its demotion from a headline component.)* The causal filter's gain is robust across kernel lengths K∈{4,8,16,50} (k16 marginally best & tightest, 5-seed 0.0676 ± 0.0002; low kernel sensitivity, §5.4). Every capacity-*adding* probe we tried instead (continuous time-decay kernel, expert heads, text-distillation, dual-text, EMA, CL4SRec, James–Stein shrinkage [Efron and Morris 1973; James and Stein 1961], niche-competition loss, heat-kernel label smoothing, spectral-shrink, cue-fusion, forced ID→text routing) was neutral or harmful — several with a learned scalar the model itself drove to zero (the negative-result map, tabulated in Table S1, §5.5).

Table 1a: SASRec-family text baselines (protocol-parity check, full-catalog eval).

Method	NDCG@10	HR@10	Notes
popularity	0.0125	0.0248	trivial floor (traceable: `results_5core_Video_Games.json`)

The v1-era SASRec/SBERT/BLaIR baseline rows and the observations derived from them have been REMOVED per the artifact-graph gate: their per-seed artifacts were not retained and the values are untraceable. Protocol parity is established *independently by the dataset-identity statistics (exact user/item match, ±1 interactions on every category), the frozen split hashes, and the per-category floor runs recorded in the pre-declaration results files.*

Table 1b: Reported and locally audited HSTU-BLaIR comparator evidence [Liu 2025, arXiv:2504.10545]:

The HSTU-BLaIR paper reports on AR2023 Video_Games 5-core LLOO with identical dataset stats to ours (25,612 items / 94,762 users / 814,585 interactions). They train for **100 epochs** following Zhai et al.'s HSTU protocol. Their single-seed numbers:

Method	NDCG@10	HR@10	Comparison to ours
SASRec [Liu 2025, single seed, 100 epochs]	**0.0573**	0.1028	−15% below our full model 0.0673 ± 0.0003
HSTU [Zhai et al. 2024]	0.0741	0.1315	+10% above our full model
HSTU-OpenAI (TE3L)	0.0742	0.1328	+10% above our full model
HSTU-BLaIR [Liu 2025]	**0.0760**	**0.1353**	**+13% above our full model; stronger reported reference**
HSTU-BLaIR local SM120 compatibility port (this audit)	0.07382 final full-eval (exported artifact; the higher best-epoch reading survives only in an unretained WSL log and is excluded from the artifact graph)	0.13234 final	stronger than ours; not a faithful pinned-environment reproduction

Our full model (0.0673) exceeds their published SASRec (0.0573) but does not reach their HSTU (0.0741) or HSTU-BLaIR (0.0760) on Video_Games. Our local compatibility-port HSTU-BLaIR run reached final full-eval NDCG@10 `0.07382` (the best-epoch reading exists only in an unretained WSL log and is excluded from the artifact graph); the run is stronger than SASRec-SBERT but carries SM120/fbgemm validity caveats.

Relative to HSTU-BLaIR, this paper contributes multi-seed internal ablations, a released preprocessing path, and implementation diagnostics—not a leaderboard result. The reported 0.0760 remains the stronger Video_Games reference. Core-block equality at one aligned configuration (§3.2) and environment-caveated local regenerations (§5.6) do not constitute a pinned end-to-end reproduction.

Older published baselines on different protocols (NOT directly comparable; for context only):

Method	NDCG@10	Dataset	Protocol
SASRec (in TIGER paper)	0.0318	Amazon Beauty **2014**	their 5-core (different category, different dataset year)
TIGER [Rajput et al. 2023]	0.0384	Amazon Beauty **2014**	their 5-core
SASRec (in LIGER paper)	0.02179±0.00023	Amazon Beauty **2014**	their 5-core
LIGER (K=20, Yang et al. [2024])	0.04020±0.00044	Amazon Beauty **2014**	their 5-core
LIGER (K=N, full retrieval)	0.04738±0.00151	Amazon Beauty **2014**	their 5-core
Best LLM-encoder in BLaIR (Hou et al. [2024]) Table 8	0.0138	AR2023 Video_Games	**by-timestamp 8:1:1, no k-core**, UniSRec downstream

Comparability caveat: TIGER and LIGER evaluate on the older Amazon Reviews 2014 dataset, not AR2023. The BLaIR paper does evaluate on AR2023 Video_Games but uses a by-timestamp split (no k-core filter) and a different downstream architecture (UniSRec), producing numbers ~3-4× lower than our 5-core LLOO numbers. The 5-core filter guarantees ≥5 interactions per user and item in the full filtered sequence (which substantially eases the benchmark relative to the by-timestamp variant); note that after the leave-last-out split a small set of items retains zero *training* occurrences (§5.3 cohort disclosure). We therefore do **not** claim SOTA over TIGER/LIGER/BLaIR. HSTU-BLaIR is the relevant stronger AR2023 Video_Games 5-core reference and blocks a SASRec-SBERT SOTA claim.

Concurrent protocol landscape. SID-MLP [Guo et al. 2026, arXiv:2605.12617], Latte [Hou et al. 2026a, arXiv:2605.06331], and GrIT [Shyam et al. 2026, arXiv:2602.19728] report same-statistics AR2023 LLOO results; ReSID [Liang et al. 2026, arXiv:2602.02338] and ChronoSID [Huang et al. 2026, arXiv:2607.03918] use a different filtered universe; *Augment or Not?* [Huang et al. 2025, arXiv:2505.23053], DiffuReason [Jiang et al. 2026, arXiv:2602.09744], and SILLM4Rec [Wu et al. 2025, MMAsia 2025] use non-interchangeable filtering or sampled-ranking setups; UniSGR [Sun et al. 2026, arXiv:2607.04068] uses private logs; and DIGER [Fu et al. 2026, arXiv:2601.19711] and ACERec [Xia et al. 2026, arXiv:2602.13573] address related semantic-ID problems outside this exact public protocol. We cite these works for coverage and make **no comparison against concurrent preprints**. HSTU-BLaIR remains the published external anchor for the counted point-estimate comparisons (§5.2).

Within our own full-catalog runs, the 11.6M-parameter model trains in about 10 minutes on one consumer GPU and reaches ≈5.4× the popularity floor (0.0125). These are efficiency and sanity-check observations, not priority or SOTA claims.

5.2 Canonical causal FIR isolation and cross-category transfer (legacy-package attribution caveat §3)

The single most important robustness result for the filter is that it transfers to a *second* category, and does so *more strongly than label smoothing*. On Musical_Instruments (a sparser AR2023 category), the full V2 stack reaches **NDCG@10 = 0.0413 ± 0.0005 (5-seed)**, **+7.9%** over a matched HSTU+TAPE base (0.0383 ± 0.0004) and **+57%** over a plain ID-only SASRec (0.0264). A two-arm comparison (not a controlled isolation — §3 initialization caveat) (best-by-val, vs the MI SBERT+TAPE base 0.0383, four 20-epoch seeds 20260609–12) separates the two supported regularizers at the arm level (bundled arms; no component isolation):

Table 1c: Musical_Instruments arm-by-arm comparison (NDCG@10, best-by-val, vs MI SBERT+TAPE base; shares are of the 5-seed V2 combined lift +0.0032).

Configuration	NDCG@10	Δ vs base	share of combined lift
MI SBERT+TAPE base (four 20-epoch seeds)	0.0383	—	—
+ label smoothing only (5-seed)	0.0391 ± 0.0001	+0.0008	26%
+ FIR package arm (filter component; attribution open) (k16, 5-seed)	**0.0408**	**+0.0025**	**77%**
+ both = V2 (k16, 5-seed)	0.0415	+0.0032	100%

(Recomputed best-by-val from the released per-seed artifacts. Against the headline 5-seed V2 stack the package arm's share is 77% (k16) / 82% (k8). LS-only is the 5-seed family `results_MI_lsonly_seed{08-12}` (0.03914 ± 0.00013); base is the four e20 seeds 20260609–12.)

The package arm contributes ≈3× what label smoothing does on the second category, and ≈80% of the combined lift in that same-seed decomposition — a package-level share, no component isolation — (77%

vs the k16 stack, 82% vs the k8 headline stack) — i.e. the historical package's gain appears on both development categories while label smoothing's does not (no formal component-by-category interaction was tested). Its kernel-length insensitivity also appears on MI (MI k8 0.0413 ± 0.0005 ≈ k16 0.0415 ± 0.0002). Component attribution for these historical runs remains open; the matched-initialization studies below provide cleaner internal contrasts, subject to their outcome-visibility limits.

Pre-declared matched-arm FIR study (`PREREG_FIR_V3.md` / E-A; mechanically adjudicated verdict W-POS; outcome-visible split). A nonsingular reparameterization, $y = x + \mathrm{conv}_\Delta(x)$ with $\Delta = 0$, made the treatment identity-initialized and gradient-active; three arms by eight seed blocks shared one verified initialization within each block. Learned FIR minus identity is **+0.002265, ordinary paired 95% CI [+0.001928, +0.002602]** on Musical_Instruments. This is a cleaner within-configuration contrast than the historical package arms, but the category and split had already informed development and the execution record contains the provenance deviations documented in the audit; it is therefore exploratory internal evidence, not fresh confirmation. The zero-FIR-weight-decay sensitivity is A2–A1 **+0.000010 [-0.000339, +0.000360],** $p = .95$: no difference was detected, but neither equivalence nor exclusion of a weight-decay pathway is established. Artifacts: `results_MI_FIRV3_{arm}_seed{seed}.json` (24 runs) and `fir_v3_adjudication.json`.

Pre-declared canonical reparameterization breadth (`PREREG_FIR_CANONICAL_BREADTH.md`; verdict `CANON-BREADTH-POS`; outcome-known/test-exposed). With one frozen K=16, 20-epoch configuration, zero category tuning, and exact per-seed matched initialization, learned minus identity is **+0.002110, ordinary paired 95% CI [+0.001820, +0.002399]** on Industrial_and_Scientific (paired $t = 17.23$; Holm-adjusted $p = 5.44 \times 10^{-7}$; 8/8 positive) and **+0.006150 [+0.005849, +0.006450]** on CDs_and_Vinyl ($t = 48.39$; Holm-adjusted $p = 8.42 \times 10^{-10}$; 8/8 positive). Supportive registered endpoints point in the same direction: HR@10/MRR differences are +0.003474/+0.001819 on Industrial_and_Scientific and +0.010728/+0.005033 on CDs_and_Vinyl; the broader registered cutoff family was not emitted by the trainer and is not claimed. The categories were selected after favorable legacy-package outcomes and TEST was evaluated each epoch, so these estimates are **descriptive robustness of the canonical parameterization, not independent confirmation or population transfer**. The intervals are ordinary paired-t intervals; Holm adjusts decisions/p-values, not confidence limits. All 32 source JSONs and the adjudication record are now hash-bound in the public artifact inventory.

Pre-declared six-arm active-control study (`PREREG_FIR_CONTROLS.md`; eight matched seed blocks; sealed one-shot final TEST; frozen verdict `CTRL-ACTIVE-CONTROL-SUPPORTED`). Every temporally active arm improved identity after Holm adjustment: learned **+0.002116 [+0.001910, +0.002322]**, fixed MA **+0.000712 [+0.000509, +0.000915]**, fixed HP **+0.000708 [+0.000510, +0.000907]**, channel-shared **+0.002197 [+0.002008, +0.002386]**, and parameter-matched nonlinear causal mixing **+0.001912 [+0.001657, +0.002166]**. Learned taps exceed fixed MA by **+0.001404 [+0.001085, +0.001723]** and fixed HP by **+0.001407 [+0.001089, +0.001726]**, but the two fixed arms are algebraically redundant scalar/sign parameterizations and do not count as independent mechanism controls. A learned-specific advantage is not established over the shared arm (**-0.000081 [-0.000337, +0.000175]**) or nonlinear arm (**+0.000204 [-0.000038, +0.000446]**); retained tests are not equivalence. This supports **causal temporal mixing/activity**, not unique necessity of learned per-channel FIR coefficients. The study was designed after the learned–identity outcome was known, ran from a dirty evolving tree, and its sidecars were not independently custodied because of a filename-suffix ignore error; we therefore classify it as outcome-known exploratory mechanism evidence despite its frozen analysis and sealed evaluator. Human pre-adjudication visibility remains an author-verification item.

Pre-declared breadth (two further categories). Under `PREREG_FIR_BREADTH.md` (committed before any run; fresh never-inspected seeds 20260713–17; mechanical adjudication in `FIR_BREADTH_RESULTS.md`), the filter was transplanted with **zero per-category tuning** and tested as a same-seed 5-seed filter-vs-no-filter contrast — an internal contrast with no comparator involved — on two categories never used in its development. **Both categories fired the frozen decision rule**: Industrial_and_Scientific same-seed Δ = **+0.0024 ± 0.0005** (pre-declared paired-analysis 95% CI [+0.0018, +0.0030]; 5/5 same-seed differences positive) and CDs_and_Vinyl same-seed Δ = **+0.0057 ± 0.0006** (pre-declared paired-analysis 95% CI [+0.0049, +0.0064]; 5/5). **Analysis-premise correction (2026-07-19):** the frozen rule *as written* is a paired Student-t, but its pairing premise is now known to be false — same-numbered seeds are not initialization-paired (§5.3 randomization disclosure) — so the pre-declared intervals are reported as executed-as-frozen outputs whose *paired interpretation is withdrawn*; the **primary supported statement** is the post-hoc independent-arm Welch analysis (we no longer call it conservative: with correlated arms neither interval has a general conservative ordering, and here the Welch intervals are in fact narrower than the paired ones), under which both effects remain positive with 95% CIs excluding zero (Industrial_and_Scientific [+0.0019, +0.0029], CDs_and_Vinyl [+0.0050, +0.0063]; graphed as `firb.*.welch`, evidence-class exploratory since the Welch analysis is post-hoc). Per the frozen claim wording as narrowed by this correction, each result is exactly: *the causal-FIR-filter arm's 5-seed improvement over the no-filter arm on that category is positive with a 95% CI excluding zero* — an internal same-seed contrast of the FIR-plus-initialization/optimizer package; nothing broader, no SOTA language, no comparator statement. The filter package is thus multi-seed-supported on **four categories**, two of them under this pre-declared breadth campaign — and now **estimated under the valid independent-arm TFV2 pre-declaration (outcome-visible; not confirmatory — §5.3(vii))** (below), which supersedes this campaign's withdrawn paired interpretation as the primary FIR evidence.

TFV2 independent-arm replication (pre-declared, outcome-visible; adjudicated 2026-07-20). The `PREREG_TAIL_FIR_V2.md` campaign (Git-committed frozen rules; outcome-visible — OTS chronology in §5.3, disclosure (vii)) reran both breadth categories as fully independent 8-vs-8 arms (fresh never-inspected seeds; no pairing premise; per-run sidecars released): Industrial_and_Scientific overall Δ = **+0.002131** (t = 16.99, 95% CI [+0.001862, +0.002400]) and CDs_and_Vinyl overall Δ = **+0.005770** (t = 26.12, 95% CI [+0.005275, +0.006266]) — **both PASS under Holm** (one family with the tail endpoint E1, §5.3). This replaces the withdrawn paired interpretation with a valid pre-declared independent-arm design; the treatment remains the FIR-plus-initialization/optimizer package. Descriptively, the filter's positive-frequency-tail improvements are +0.000775 (IS, $p = 2\times10^{-4}$) and +0.002329 (CDs, $p = 3\times10^{-12}$).

Comparator ablations (audit-requested). Each arm targets one named design element of the filter (with the disclosed caveat that the singular headline initialization means these arms also alter the optimization path, §3) on the same 5-seed MI stack: freezing the kernel at a causal moving average (learnable gate retained) recovers only +0.00133 of the filter's +0.00225 gain over the no-filter baseline (0.04046 ± 0.00022 vs 0.04138 ± 0.00053; non-overlapping bands) — an observation confounded by the shared singular start (§3): no learned-shape attribution is claimed. Freezing the gate at 1 (delta-initialized kernel) matches the full filter (0.04127 ± 0.00046), so the zero-init gate comparison is confounded by the singular-initialization bootstrap (§3 disclosure) and is not a clean single-factor attribution.

Pre-declared confirmation vs the published per-category reference. The seeds above (20260608–17) are *exploratory*: kernel choice, epoch budget, and the decision to compare against the published Musical_Instruments HSTU-BLaIR number (0.0406; Liu 2025, Table 2) were all made while inspecting them. To make the comparison selection-free, we froze a version-controlled pre-declared protocol (`SOTA_CONFIRM_PREREG_V2.md`, committed to version control before any confirmation run: exact config and commands, data SHA256s, decision rule, claim wording) and ran **five**

pre-specified never-inspected seeds (20260618–22) under each of the two kernels — ten arm-runs with per-run provenance manifests (command, git commit + clean-tree flag, environment, data hashes) and per-user record sidecars. Result: **K=16 fresh 5-seed 0.04152 ± 0.00045 (95% CI lower bound 0.04096) and K=8 0.04120 ± 0.00030 (CI-LB 0.04083) — both CI lower bounds above the published 0.0406, 10/10 seeds above.** The supported claim is exactly this: *fresh multi-seed means and seed-level confidence intervals exceed the published HSTU-BLaIR point estimate on this category under our reproduced protocol (parity evidence: users/items match the paper exactly; interactions differ by one — see* `SOTA_CONFIRM_PREREG_V2_ERRATA.md` *E1; our plain SASRec floor is* below *their published SASRec, ruling out an easier split).* The comparator is single-seed; our post-hoc local regeneration of it (§5.6 — the reference implementation's research path under data-movement shims, unpinned environment) regenerates the published value at its best full-eval epoch (0.0406; final epoch 0.0391) but is itself a single environment-caveated run, so no paired or distributional superiority is claimed, and this remains a per-category result — Video_Games is explicitly not claimed (§5.1). The first execution of this confirmation was voided by its own provenance tripwire (a concurrent documentation edit dirtied the tracked tree mid-campaign) and re-executed in full under a clean tree; both executions agree per-seed to ±0.0003 (`SOTA_CONFIRM_V2_RESULTS.md`). One protocol deviation is disclosed rather than claimed away: the gated runs executed at a documentation-only descendant commit of the pre-declaration's introducing commit (the pre-declared commit-equality rule was not literally satisfied); code identity across the two commits is demonstrated by empty protocol-file diffs and, in the from-scratch rebuild, by code hashes embedded in each run manifest (`SOTA_CONFIRM_PREREG_V2_ERRATA.md`, E3).

§5.2 (regenerated): pre-declared dual-kernel V2 confirmation (SOTACONF_V2, fresh seeds 20260618–22, EXEC2 gated artifacts, n_eval=57,439).

kernel	fresh 5-seed NDCG@10	95% CI lower bound	vs published 0.0406
K=16	0.04152 ± 0.00045	0.04096	ABOVE
K=8	0.04120 ± 0.00030	0.04083	ABOVE

Fresh seeds above 0.0406: 10/10. EXEC1↔EXEC2 max per-seed $|\Delta|$ = 0.000057 (< 0.0003). Clean rebuild (rebuild_v2/): K=16 CI-LB 0.04092, K=8 CI-LB 0.04084 — dual gate PASS.

Office_Products V1 (the original second-category attempt) — descriptive only; superseded by the counted V3 campaign below. Office_Products numerically exceeded the published HSTU-BLaIR point estimate under the frozen MI configuration, but the pre-declared SASRec floor check failed because our local SASRec floor was substantially above the published SASRec reference. We therefore treat the V1 Office campaign as descriptive evidence, not as a passed confirmatory category (the separately pre-declared V3 campaign later passed and is counted, §5.2). The floor anomaly has since been resolved mechanistically — the reference implementation's own SASRec, run locally on its own pipeline (§5.6), lands +13.9% above its published row, while a completed run of their Office HSTU-BLaIR configuration regenerates its published row (+1.6%; §5.6) — the conservatism is a property of the published Office SASRec row specifically. The VOID is retained on procedural grounds (the pre-declared floor check failed as written); full detail, decomposition, and disclosures in Appendix A.0.

Office_Products V3 (redesigned pre-declaration) — PASSED. Under `PREREG_OFFICE_V3.md` (committed before any run; fresh never-inspected seeds 20260728–32; gate: each arm's fresh 5-seed 95% CI lower bound must exceed the **environment-matched local regeneration of the comparator** — best full-eval 0.0279, itself above the published 0.0271 — with ≥4/5 seeds individually above), **both kernel arms passed**: K=16 fresh 5-seed **0.03047 ± 0.00011 (CI-LB**

0.03033) and K=8 **0.03029 ± 0.00005 (CI-LB 0.03024)**, 10/10 seeds above both references; dataset identity, reference-artifact hashes, and per-run provenance verified mechanically (`OFFICE_V3_RESULTS.md`). Per the frozen claim wording: *a per-category point-estimate comparison against the published 0.0271 and its environment-matched single-run local re-generation 0.0279; no paired or distributional superiority is claimed; this is not SOTA on Office_Products, not SOTA on Amazon Reviews 2023, and not a general-SOTA claim of any kind.* The V1 VOID above stands unchanged as its own record (Appendix A.0).

5.3 The Musical_Instruments frequency-5 tail case (cross-dataset heterogeneity not established)

Beyond *whether* text helps overall (modestly: +2.7% on Video_Games, §5.1), we ask *where this additive frozen-text construction helps* (only early additive fusion of frozen text features is tested — §2.3 scope note). `evaluate()` reports `by_popularity` NDCG@10 over nominal train-frequency terciles {tail, mid, head} (bucketed by TRAIN frequency only, so no validation/test frequencies enter the bucketing; two cohort-definition defects are disclosed below), and we compare the full text stack against an ID-only ablation on the **same split and seed numbers**. The cold-start intuition — text should rescue rare items where ID embeddings are starved — holds on *one* of three datasets, and the MI difference survives an independent-arm analysis.

Randomization disclosure. Same-numbered text and ID arms are not initialization-paired: treatment-specific modules consume RNG before the shared backbone is constructed, changing backbone initialization and subsequent stochastic streams. Table 1d's per-seed differences are therefore descriptive, all retained inference is independent-arm Welch/Satterthwaite, and the former paired CI/MDE/TOST analyses are withdrawn.

Table 1d: text − ID tail-tercile NDCG@10 contrast (same-seed-number arms — NOT initialization-paired; per-dataset estimates — cross-dataset heterogeneity not established, point 3).

dataset	catalog density	tail Δ NDCG@10 (5-seed)	seeds positive	verdict
Musical_Instruments	sparse	**+0.000335 ± 0.000195**, Welch 95% CI [+0.000109, +0.000562] **excl 0**	**5/5**	**frequency-5-heavy tail advantage (independent-arm robust; §5.3)**
Video_Games	dense	−0.000148 ± 0.000179	2/5	null — equivalence **NOT** established (retraction, point 2)
Beauty_and_PC	dense	−0.0000078 (3-seed, sample-sd 0.000020)	1/3	exploratory (3-seed; mixed eval geometry)

The historical MI contrast survives independent-arm analysis (Welch $t = 3.94$, $p = 0.014$, 95% CI [+0.000109, +0.000562]) and has tail HR Δ = +0.00109 ± 0.00065, but its absolute scale is small (≈29.6 versus 20.0 hits/seed); MI's largest absolute text lift is at the head (+0.00527). On Video_Games, no tail difference is detected (Welch $p \approx 0.22$), but equivalence is **not** established. Beauty remains exploratory because one arm pair uses unmatched evaluation geometry. Most importantly, the repaired TFV2 MI–VG contrast is +0.000247 ($p = 0.13$), so cross-dataset heterogeneity did not replicate and is not claimed (Fig. 1).

Why Table 1d is historical. Its nominal terciles mix zero-train-exposure targets with positive-frequency targets and split boundary-frequency ties by item-ID order; the original runs also lack per-row outcome sidecars. These arm-independent defects do not create a treatment-specific cohort, but they prevent the intended rare-item interpretation. TFV2 therefore separates zero exposure, uses whole-frequency-group cohorts, and releases per-user outcomes.

Repaired-estimand campaign (TFV2; 8 independent seeds/arm). On the tie-safe MI positive-frequency tail, text − ID is **+0.000420, 95% CI [+0.000181, +0.000660], Welch p = 0.0022**, passing the frozen Holm rule. The [1,6] band (+0.000546, p = 0.0039) and HR@10 (+0.001150, p = $4{\times}10^{-5}$) agree. However, zero-exposure targets produce no hits through rank 100 in any of four categories, the exclude-frequency-5 sensitivity is nonsignificant (+0.000071, p = 0.52), and the frequency-5 group alone is +0.001379. We therefore report a **frequency-5-heavy positive-tail case**, not cold-start capability or a smooth rare-item law. VG is nonsignificant (+0.000173, p = 0.14), as is MI–VG (+0.000247, p = 0.13).

Evidence status. TFV2's endpoints were Git-frozen before launch, but interim test values became visible, its first independently verifiable timestamp postdates the first result, and the adjudicator was committed after some outputs existed. The eight-seed size was resource-based rather than prospectively powered, and the run manifests record concurrent tracked-document edits. TFV2 is therefore **outcome-visible, not confirmatory**; its released sidecars and frozen estimator support reproducible estimates, not a claim of blinded confirmation. The repository retains the complete chronology, cohort files, logs, hashes, and errata.

5.4 Titration: thinning tracks the head effect but not the tail effect (level contrasts under a bundled intervention; head trend exploratory and uncorrected; tail inconclusive)

We thinned Video_Games interactions over six rungs down to MI's global density (ρ=0.66), holding n_eval at 94,762 and re-estimating text–ID differences with five seeds per rung. The two strata respond differently:

- **Head:** Δ rises overall from +0.00221 to +0.00354; all rungs are 5/5 positive. Spearman ρ_s versus density is -0.94 (exact p=0.017), with one adjacent reversal; the HR trend is weaker (ρ_s=-0.71, p≈0.14). This is an exploratory uncorrected rank trend on one fixed thinning draw.
- **Tail:** Δ is non-monotone (ρ_s=-0.14). At MI-matched density it is -0.000108 ±0.000503 (1/5 positive), so thinning does not reproduce MI's +0.000335 tail estimate. Without a prospective equivalence margin this is inconclusive, not evidence of no effect.

The full ladder is Table S2. The head trend and tail non-reproduction are level contrasts under a bundled intervention; their interaction is nonsignificant. Global density is therefore not identified as the tail mechanism.

5.4.1 The level contrasts in one scale-free table: the cross-dataset tail/head arm ratio (descriptive; no interaction-test support). To separate absolute scale from arm contrast, Table 5 reports the text/ID NDCG ratio at full VG density, VG thinned to MI density, and native MI. Ratios above one favor text.

regime (interactions/item)	n (text/id)	TAIL text	TAIL id	**TAIL ratio**	tail Δ	**HEAD ratio**
VG full density (ipi 24.5)	5 / 5	0.004919	0.005068	**0.971 (−2.9%)**	−0.000148	1.026 (+2.6%)
VG thinned → MI-density (ρ=0.66, ipi 16.2)	5 / 5	0.003569	0.003677	**0.971 (−2.9%)**	−0.000108	1.049 (+4.9%)
MI native (ipi 16.2)	5 / 5	0.001551	0.001216	**1.276 (+27.6%)**	+0.000335	1.101 (+10.1%)

The head ratio rises from 1.026 to 1.049 as VG is thinned, toward MI's 1.101. The tail ratio remains 0.971 at both VG densities, far below MI's 1.276. Thus matching global density does not reproduce the MI tail advantage on this draw. The earlier two-seed "trough" interpretation is retracted; the five-seed endpoint is near zero with a wide interval. This remains a level contrast, not a significant stratum-by-density interaction.

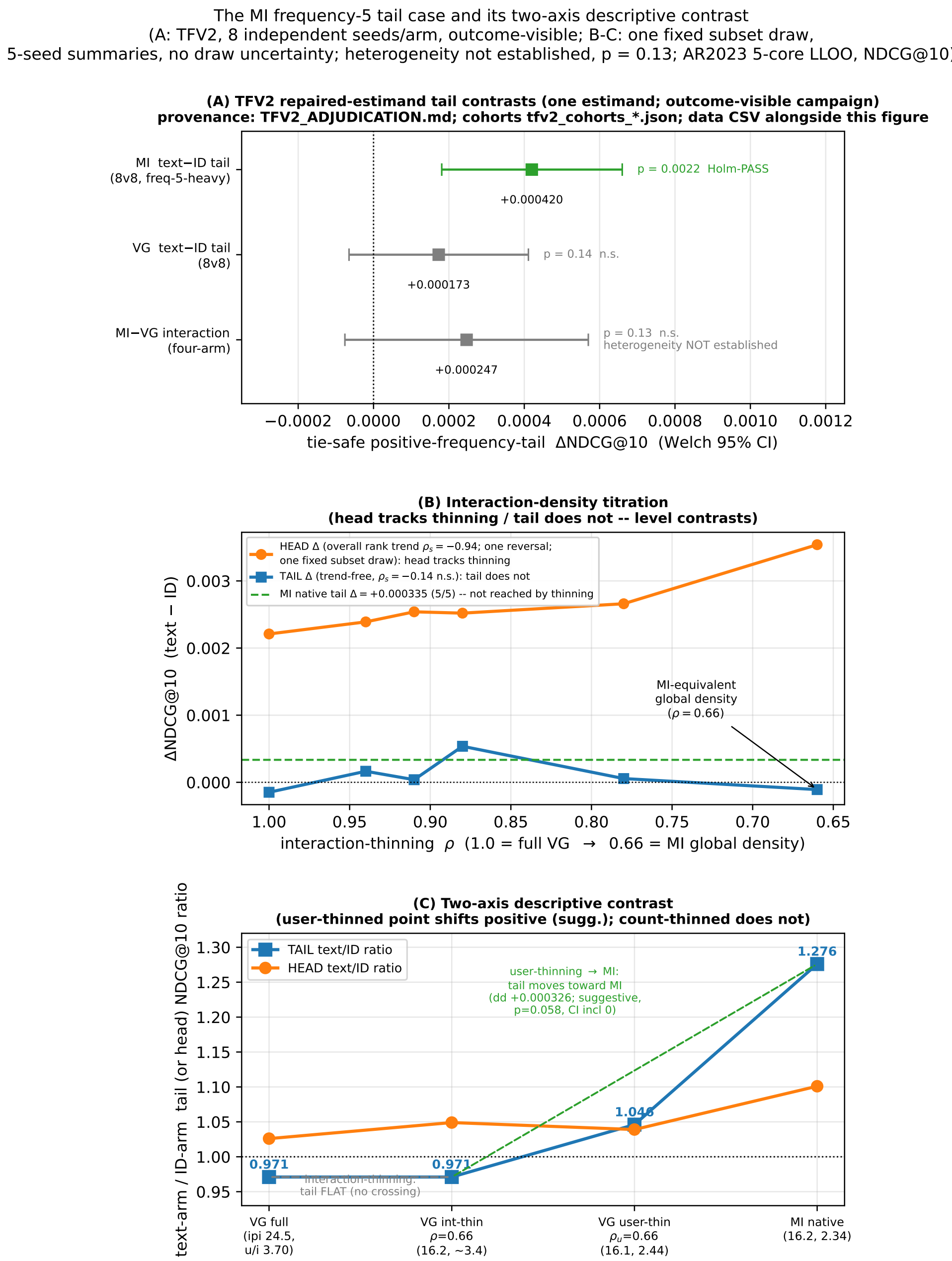

Fig. 1. (A) TFV2 repaired-estimand tail contrasts — MI, VG, and the MI−VG interaction as Welch 95% CIs (8 fresh seeds per arm; one estimand; outcome-visible campaign, §5.3; cross-dataset heterogeneity not established, p = 0.13). The historical defective-cohort Table 1d values are tabulated, not plotted. (B) Interaction-density titration level contrasts and (C) two-axis descriptive ratios — both on ONE fixed subset draw with no draw/seed uncertainty shown; arms are not initialization-paired (§5.3); the panel-C shift is suggestive only (p = 0.058, CI includes 0). Plotted values, estimator, and analysis IDs: figures/fig_tail_law_mechanism_data.csv.

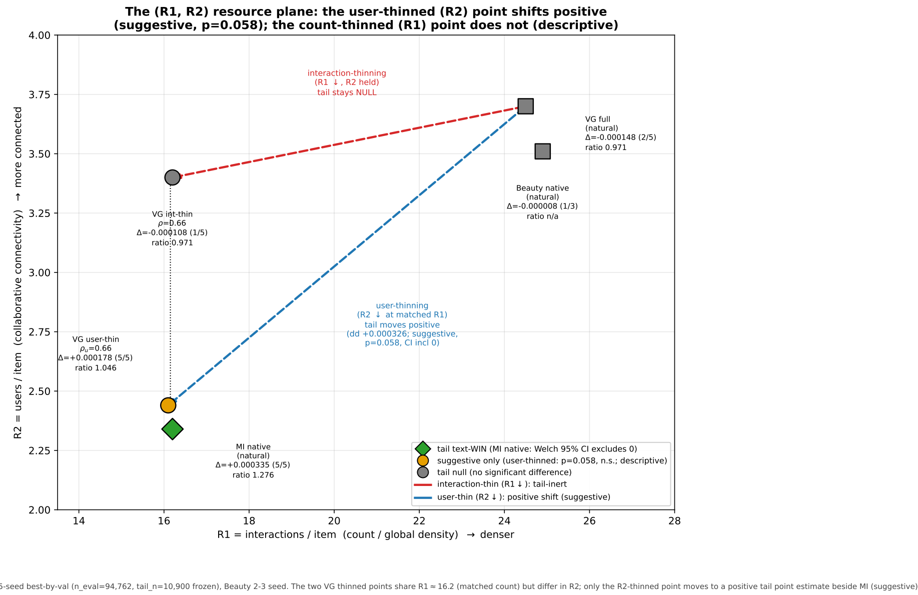

Fig. 2. The (R1 = interactions/item, R2 = users/item) resource plane — the two VG thinned rungs share R1, but only the user-thinned (R2) point moves to a positive tail point estimate beside MI native (suggestive, p = 0.058; descriptive contrast; bundled interventions on one fixed subset draw; the §5.4.2 dd inference is suggestive, not significant, under the corrected independent-arm analysis).

One hypothesis is that user connectivity or item-text quality differs between native MI sparsity and artificially thinned VG. Neither is identified here: user thinning changes several dataset properties at once, and an item-text permutation control has not been run.

5.4.2 User-mode titration: a suggestive tail-level response at MI-matched connectivity (bundled intervention; descriptive). User thinning preserves each retained user's history while lowering users/item. At ρ_{user}=0.66 it matches the interaction-thinned rung's interactions/item (16.1 versus 16.2) and approaches MI's users/item (2.44 versus 2.34). Because user composition, degrees, update counts, and topology also change, this is a bundled-scheme contrast, not single-factor identification.

The read comes out on the predicted branch — a positive shift (suggestive):

regime (int/item, users/item)	TAIL ratio	tail Δ (5-seed)	HEAD ratio
VG full (24.5, 3.70)	0.971	−0.000148	1.026
VG interaction-thinned ρ=0.66 (16.2, ~3.4)	0.971	−0.000108 ± 0.000503 (1/5 pos)	1.049
VG user-thinned ρ_{user}=0.66 (16.1, 2.44)	**1.046**	**+0.000178 ± 0.000153 (5/5 pos, t≈2.6)**	1.039
MI native (16.2, 2.34)	1.276	+0.000335 ± 0.000195 (5/5 pos)	1.101

The tail ratio moves from 0.971 to 1.046. The independent-arm difference-in-differences is +0.000326 (p=0.058, 95% CI [-0.00001, 0.00067]); the within-rung tail level is +0.000178 (p≈0.16). The former paired analysis is retracted because arms are not initialization-paired. Fig. 2 therefore shows a suggestive direction, not a confirmed effect or connectivity attribution.

Lower user-thinning rungs do not support a "peak then collapse" interpretation: ρ_{user}=0.50 is +0.000236 ±0.000532, while ρ_{user}=0.40 exhibits a text-arm floor collapse. The user-thinned estimate also remains below native MI. Overall, interaction thinning tracks the head level, while user thinning gives only a suggestive tail-level shift. Shared intervention draws omit subset uncertainty, the stratum-by-density interaction is nonsignificant, and the mechanism remains unidentified. A prior spectral explanation is retracted because it did not measure the claimed spectrum [Baik et al. 2005; Gavish and Donoho 2014; Marchenko and Pastur 1967].

> **Provenance:** released `results_USERTITR_*.json`; five seeds per rung; n_eval=94,762 and tail_n=10,900 fixed. Arms share seed numbers but are not initialization-paired. All model seeds use one intervention draw.

5.5 Screening log of capacity-adding probes (a search record, not powered exclusions)

A central empirical finding of this work is **asymmetric**: the only levers that moved test NDCG@10 were *capacity-restricting* (label smoothing, the causal FIR filter) or classical inductive biases already in Table 1 (the time bias); **no capacity-*adding* probe we tried showed a benefit at its tested power (mostly single-seed).** We document the full set here as a first-class result — negatives are part of the contribution, and several are *interpretable* negatives in which a learnable scalar, free to engage the mechanism, was driven by the optimizer to (or past) zero. This is observational evidence — most rows are single-seed probes (power note below) — that the in-environment ceiling was not left unexplored on these axes; the residual gap to the published HSTU-BLaIR 0.0760 could not be isolated (the pinned reference environment is not installable here, and the shimmed local execution of §5.6 did not include the Video_Games configuration); what the probe set shows is that none of the tested probes (mostly single-seed; Table S1 power note) produced a gain — observations that leave the gap unexplained, not exclusions.

All entries are best-by-val test NDCG@10, full-catalog eval (n_eval = 94,762 on Video_Games), on the same fixed protocol. Δ is versus the stated base; for the three learnable-scalar probes (W1/X1/Y1) we report the absolute single-seed test next to its base band, since the headline test is statistically indistinguishable from the base and the *learned scalar* is the load-bearing readout. Several rows are single-seed — sufficient for a negative-result map (a probe that fails to clear its base on one seed, or whose own learned scalar switches it off, does not warrant five seeds).

Unequal power, stated plainly. The probes in Table S1 do *not* carry equal evidential weight: most rows are single-seed exploratory probes, while a few were run at full multi-seed power. Only the multi-seed, pre-declared negatives — the conn-gate five-seed confirmation (cadence-caveated; see the row note) and the titration nulls of §5.3–§5.4 (the VG tail null — former equivalence claims retracted, §5.3 — and the 5-seed-locked ladder rungs) — were run at full pre-declared power (no null is thereby confirmed); the **Beauty near-zero tail estimate is a 3-seed exploratory observation** (mixed evaluation geometry, §5.3) and carries **no** confirmatory weight. The remaining rows are exploratory documentation of the search space, recorded for completeness and interpretability, not powered hypothesis tests. Full per-probe seed counts are given in the table's n column.

The full screening log — all 18 capacity-adding / alternative-mechanism probes with per-probe Δ NDCG@10, seed counts, and learned-scalar readouts — is **Table S1** (Supplement §S.4); it is summarised above and analysed next.

The pattern is consistent and is itself the finding: across loss-side (W1, Y1, sampled-softmax), embedding-side (X1, GD1, text-init), attention-side (c3, n_heads, text-sim), encoder-side (BLaIR, dual-text), and auxiliary-objective (c1, c2, CL4SRec, EMA) axes, no capacity-adding probe produced a test gain at its tested power (mostly single-seed; power note above), and three free scalars (W1's β, X1's c, Y1's T) were each switched off — or, for W1, reversed — by the optimizer. Only anti-overconfidence regularizers (label smoothing) and the restricted causal-filter inductive bias (Table 1) transferred. These are hypothesis-generating observations at their stated (mostly single-seed) power, not powered exclusions; **we do not claim they bound an in-environment ceiling**. (An earlier spectral-irreducibility clause here is retracted — see the §5.4.2 retraction note.) The conn-gate row deserves separate emphasis: it is the only entry tested at full five-seed power and the only one that directly operationalizes *this paper's own intervention-associated tail pattern* — the collaborative-connectivity association of §5.4.2. A single seed showed a +0.00053 tail lift, but the pre-declared five-seed confirmation collapsed to +0.0001 ± 0.0003 (95% CI [−0.00025, +0.00045], includes zero), and the free gate scalar α was driven *below* its initialization (voted off; the learned value is recorded in 4/5 seed logs — the fifth predates the log line, and the graph reports the recovered count). **Cadence caveat (disclosed 2026-07-19):** the conn-gate arms were evaluated only at epochs {10, 20} while their base evaluated every epoch, so best-by-val checkpoint opportunities were unequal across arms; the tail CI-includes-zero verdict is unchanged by this, but the small overall delta is cadence-sensitive and we rest no claim on its sign. This is the cleanest illustration of the paper's central distinction: a mechanism-associated pattern can be *supported* by controlled thinning contrasts (the matched-R1 level contrast) yet not be *actionable* through a learnable cold-start gate — one *interpretation* — a hypothesis, not an identified mechanism — is that the connectivity signal the model needs is already latent in the frozen text features, leaving no residual for an explicit gate to exploit.

5.6 Executing the reference implementation locally (post-hoc comparator regeneration)

Late in this work we succeeded in running the **reference implementation itself** — the HSTU-BLaIR repository at its released commit, unmodified (clean git tree): its preprocessing, its gin configurations, its trainer, and its eval protocol — end-to-end on our hardware. The pinned environment (torch 2.2.2 / fbgemm_gpu 0.6.0) remains uninstallable here (no sm_120 kernel images, no Windows wheels); what makes execution possible under our unpinned torch 2.11 environment is that the repository's research path uses exactly **three** `fbgemm` operators, all pure data movement (`asynchronous_complete_cumsum`, `jagged_to_padded_dense`, `dense_to_jagged`), which we re-implement in plain PyTorch with self-tested forward/gradient parity, plus a world-size-1 identity DDP wrapper and logging/data-acquisition patches (full shim inventory, per-run provenance, and artifacts: `THEIRS_ON_OURS_REPORT.md`). No model math is touched — the layer norms, fused projections, SiLU pointwise attention, biases, loss, and scoring all execute in the reference repository's own unmodified lines, consistent with the core-block parity evidence of §3.2. Their own preprocessing pipeline ran with its dataset-size assertions left active and **passing** on both corpora (Musical_Instruments: 24,587 items / 57,439 users; Office_Products: 77,551 items / 223,308 users — the comparator paper's exact statistics).

Two single-run results (their configurations as shipped, 101 epochs; our RTX 5060 Ti):

run (their code, their data, their eval)	published	local, final epoch	local, best full eval
MI HSTU-BLaIR — NDCG@10	0.0406	0.0391	**0.0406** (epoch 35 — exact)
MI HSTU-BLaIR — HR@10	0.0733	0.0716	0.0743
MI HSTU-BLaIR — MRR	0.0371	0.0355	0.0369
Office SASRec — NDCG@10	0.0153	**0.0174** (+13.9%)	0.0177
Office HSTU-BLaIR — NDCG@10	0.0271	0.0275 (+1.6%)	0.0279

The Musical_Instruments comparator regenerates locally. The final-epoch full-corpus eval lands within 2.3–4.3% of every published metric above, and the best full-eval epoch hits the published NDCG@10 exactly; the run oscillates in a 0.0384–0.0406 band from roughly epoch 30 onward (the comparator's own README notes run-to-run variability). The published 0.0406 that §5.2's pre-declared confirmation is measured against is therefore no longer only a transcribed constant: a local run of the generating code regenerates it under the unpinned shimmed research path, and our pre-declared CI lower bounds (0.04096 / 0.04083) sit above both its best (0.0406) and final (0.0391) local readings. **The Office_Products floor anomaly resolves**: their own SASRec configuration lands +13.9% above its published row here, crossing the published value at epoch 20 of 101 and never re-entering — which, combined with our floor run, decomposes the +44% floor-check failure entirely into published-row conservatism and baseline-strength protocol differences, leaving nothing attributable to our split or eval (resolution in Appendix A.0; the VOID is deliberately retained).

Caveats (why these are regenerations, not reproductions). Single runs (the reference code seeds only Python's `random`, so even in place it is not run-to-run deterministic); an unpinned environment (torch 2.11 vs pinned 2.2.2, transformers 5.7 vs 4.51, different cuBLAS/TF32 kernel versions); a different GPU than the comparator used; and the shims themselves — though the shim risk has since been closed on CPU: under the pinned torch 2.2.2 itself, the real `fbgemm-gpu(-cpu)==0.6.0` operators match our shims **bit-exactly (max abs diff 0.0, forward and gradient)** on every input the pinned binary accepts, the reference block under {pinned torch + real fbgemm} vs {pinned torch + shims} is bit-exact at every stage, and the remaining torch 2.2.2→2.11 block-level numerics drift is bounded at **4.8×10^{-7} per stage / 2.4×10^{-7} end-to-end**, with negative controls confirming the harness detects 1-ulp perturbations (`PINNED_ENV_PARITY_REPORT.md`); the pinned stack's *GPU* (cu121) kernel numerics remain the one locally untestable component. We therefore cite these as **environment-caveated single-run regenerations** and change no claim wording on their basis: §5.2's claim remains a point-estimate comparison against the published number.

5.7 Pre-declared fresh-seed hybrid: late fusion with a closed-form item-item model (E-F; adjudicated 2026-07-23; W-H-POS on all three categories)

Provenance and system. An author-operated side campaign (2026-07-22; outcome-visible development on historical seeds, disclosed as such) found that late score fusion of the frozen sequential stack with EASE [Steck 2019] — a closed-form linear item-item model fit on the *train split only* — improved MI test NDCG@10 on every historical seed. `PREREG_HYBRID_V1.md` (frozen and committed before launch, with its mechanical adjudicator) is the fresh-seed replication: per user, fused = z(seq) + w·z(EASE); (l2, w) selected on *validation only* from frozen grids; the fusion stage evaluates test once, at the selected pair (the underlying trainer additionally logs test curves at eval epochs under the pipeline's long-standing best-by-val protocol — a standing test exposure disclosed in §5.3 process disclosure (i), not a fusion-stage selection input); Completed per-seed fusion results were also read for operational monitoring before the full family finished; PREREG_HYBRID_V1 registered no no-interim clause, and the once-only adjudication ran after all files existed (disclosed). Five fresh seeds 20260721–25 per category; integrity gates all passed (per-seed reconstruction fidelity <0.0005 vs the recorded base result; frozen-grid membership; canonical n_eval; mechanical config equality to the per-category frozen reference), adjudicator exit 0.

Result (frozen wording; 'pre-declared' replaces the prereg's synonym for the manuscript's uniform terminology). Under the pre-declared fresh-seed replication (PREREG_HYBRID_V1, 5 seeds), late z-score fusion with a train-only EASE model changed test NDCG@10 by **+0.00244 [+0.00219, +0.00268]** on Musical_Instruments, **+0.00261 [+0.00211, +0.00310]** on Industrial_and_Scientific, and **+0.00317 [+0.00286, +0.00348]** on Video_Games (Holm-corrected paired t across the three categories; raw p = 1.0e-5 on MI, 1.3e-4 on IS, 9.0e-6 on VG, all Holm-significant;

the significance statement attaches to the fused-vs-sequential contrast, never to any published comparator). The hybrid is a two-scorer system; published comparator numbers are single-scorer, single-run values, and every comparison below remains a point-estimate comparison under the environment caveats of §3. Fresh-seed sequential-only means for context: MI 0.04155, IS 0.03340, VG 0.06713.

Published-comparator rows (frozen wordings). The fused system's five-fresh-seed mean test NDCG@10 on Musical_Instruments, **0.04399**, exceeds the published single-run HSTU-BLaIR point estimate 0.0406 under our environment-caveated regeneration protocol (point-estimate comparison; no distributional claim). On Video_Games the fused five-fresh-seed mean, 0.07031, remains below the published single-run 0.0760. A five-checkpoint ensemble of the same frozen stack, fused with EASE under the same validation-only rule, scored **0.04557** on Musical_Instruments (single evaluation; ensemble frame — comparable only to other ensembles).

Attribution and scope. EASE is Steck (2019); z-score late fusion is standard. The local contribution is empirical: the measured complementarity between a text-augmented sequential scorer and a closed-form co-occurrence scorer under the 5-core LLOO full-catalog protocol, quantified under a frozen plan on fresh seeds. EASE alone is far weaker than the sequential model on these catalogs (side-campaign observation; the gain is complementarity, not a stronger component). The dense closed-form fit is infeasible at Office/CDs/Beauty catalog sizes; that scope exclusion was declared in the prereg, and a content-based third scorer that avoids the item-item inversion was subsequently studied at five-category scale (§5.8; outcome-visible, protocol-deviated — see its classification). Artifacts: results_{MI,IS,VG}_HYBRIDV1_base_seed*.json (+.fusion.json, per-user records), results_MI_HYBRIDV1_ensemble5.json, hybrid_v1_adjudication.json.

5.8 Sparse-warm text-fusion study on all five categories (E-G; outcome-visible, protocol-deviated; descriptive estimates, no pre-declared-confirmation status; a near-threshold redistribution, not cold start)

Classification. All 25 runs completed, but the campaign is outcome-visible and protocol-deviated. Interim endpoints were exposed, the literal frozen configuration gate fails on MI and VG, and the amended gate postdates outcome visibility. The literal failure governs; the amended PASS is a sensitivity only. A planned replication was also exposed before adjudication. Accordingly, every estimate below is descriptive and conditional on optimizer seeds over one repeatedly used split.

System. At evaluation time, a history-centroid text score is added to the sequential score and, where feasible, EASE. Text weights depend only on train-frequency bins and are selected on validation under an overall non-inferiority constraint. The text scorer fits no parameters, but the sequential and EASE components are trained or fit; this is not a training-free recommender.

The repository retains the as-launched gate, later sensitivity gates, their hashes, and the exact chronology. This preserves the defect without converting a post-outcome repair into a frozen decision.

Descriptive result. Tail-bin NDCG@10 changes were +0.00217 [0.00196, 0.00239] on MI, +0.00243 [0.00218, 0.00267] on IS, +0.00337 [0.00305, 0.00370] on VG, +0.00114 [0.00105, 0.00123] on Office, and +0.00481 [0.00472, 0.00490] on CDs. Overall changes were near zero. These intervals are retained as descriptive arithmetic, not as pre-declared confirmation.

The gain is entirely frequency 1–5; frequency-zero targets never change at rank 10. Mid and head means decline in every category, so this is a sparse-warm redistribution rather than free overall accuracy. All validation choices hit the text-weight grid ceiling, leaving the optimum unbracketed. With five seeds, the exact two-sided sign-test floor is 0.0625.

Evaluation-time content fusion is established prior art [Brody and Lagziel 2024; Collins et al. 2025; Lichtenberg et al. 2025; Liu et al. 2023b; Wang et al. 2024, 2025; Xu et al. 2024]. We claim no architectural novelty for this element; it is retained only as a protocol-deviated redistribution study on fixed splits. Full operational chronology and artifacts are in the versioned supplement and repository.

6 Discussion

6.1 Interpretation

The main result is deliberately modular and now more narrowly identified. Positive matched FIR–identity contrasts on three outcome-visible categories show that activating the residual changes this backbone beneficially in the tested settings. The active-control study shows that shared and parameter-matched nonlinear causal mixing can recover the gain without learned per-channel FIR taps. The supported mechanism is therefore causal temporal mixing/activity, not uniquely learned FIR coefficients, universal frequency-filter benefit, or a new end-to-end architecture. All breadth and control studies are internal, outcome-known evidence rather than external-comparator or independent-confirmation evidence.

The supporting text studies sharpen that boundary. Frozen text features add only +2.7% overall on Video_Games, and TAPE is sub-additive (+0.0009 in one seed; +0.0004 in the four-seed check). The repaired MI analysis finds a small frequency-5-heavy benefit rather than cold-start capability: zero-exposure targets receive no hits through rank 100, the exclude-boundary sensitivity is nonsignificant, and the MI–VG contrast does not replicate. The Beauty_and_PC scan (Appendix A.1) likewise attributes its small transfer signal to the bundled BLaIR-rich-text representation, not to the MLP adaptor in isolation.

6.2 Practical implications

The FIR residual is attractive because it is small, causal, and detachable: it can be initialized as the identity and added without changing the item scorer or evaluation protocol. In this implementation, chunked full softmax makes 207k-item training feasible, cached item features substantially reduce evaluation cost, and right-padding with a causal-only mask avoids the NaN failure observed with the former masking path. Conversely, heavy dropout, removal of learned item embeddings, and simple subsequence augmentation did not transfer in the Beauty supporting study; these are observations under this pipeline, not general exclusions.

6.3 Claim limits and remaining tests

- **Comparator uncertainty.** The counted Musical_Instruments and Office V3 comparisons are against published single-run point estimates and an environment-matched single-run regeneration; no paired or distributional superiority over HSTU-BLaIR is claimed. The official end-to-end CUDA/Triton system cannot run on the local sm_120 GPU. Only the research block at an explicitly aligned configuration is numerically matched; trained-checkpoint and end-to-end equivalence are not claimed.
- **Historical, canonical, and control evidence.** Same-numbered historical arms were not initialization-paired, so their paired interpretations are withdrawn. E-A and canonical breadth use matched initialization, but reused outcome-visible splits; the latter categories were selected after favorable package outcomes and TEST was evaluated each epoch. Their ordinary paired intervals are descriptive internal robustness. The active-control final evaluation was sealed, but the campaign ran across a dirty evolving tree and its sidecars lacked independent

custody because the ignore rule covered the wrong suffix. Fixed MA/HP are algebraically redundant. Shared and nonlinear controls prevent a learned-tap-specific claim. Office V1 remains **VOID**; Office V3 supports only its frozen point-estimate wording.

- **Tail and probe scope.** The tail result is AR2023-specific, small in absolute terms, concentrated at train frequency 5, and not replicated on a non-Amazon domain. The original cohorts mixed exposure regimes and split frequency ties; TFV2 repaired those estimands but was outcome-visible and is not classified as confirmatory. Most capacity probes are single-seed screens, and nonsignificance is never interpreted as equivalence.
- **Open controls.** Item-text permutation and objective/target-multiplicity parity controls remain unrun, so the small text-stack gain cannot yet be attributed entirely to semantic alignment. The thinning studies intervene synthetically on one fixed draw and do not identify the real-world data-generating process.
- **Provenance and input scope.** One confirmation campaign ran at a documentation-only descendant of its frozen commit; code hashes and empty protocol-file diffs support code identity, but the literal commit-equality deviation remains disclosed. The canonical model consumes only item IDs, timestamps, and frozen item-text embeddings—never outputs or representations from a comparator model.

7 Conclusion

This paper contributes a narrow causal-filter module and an evaluation discipline for testing it. The discipline combines git-frozen protocols, a fail-closed artifact graph, environment-caveated comparator regeneration, and symmetric adjudication. Its value is illustrated by the Office_Products V1 campaign: its own floor check VOIDed the comparison, and that verdict remains in the record even though a redesigned protocol later passed on fresh seeds.

The main empirical result is a family of identity-initialized, gradient-active, strictly causal temporal-mixing modules. Learned FIR−identity estimates are positive on Musical_Instruments (+0.002265, 95% CI [0.001928, 0.002602]), Industrial_and_Scientific (+0.002110 [0.001820, 0.002399]), and CDs_and_Vinyl (+0.006150 [0.005849, 0.006450]), but these are outcome-visible internal estimates. In the active-control study, every temporally active arm improves identity; learned taps beat the redundant fixed-filter parameterizations but are not separated from shared or parameter-matched nonlinear causal controls. The evidence therefore supports causal temporal mixing in this configuration, not unique necessity of learned per-channel FIR coefficients, external-comparator superiority, or independent cross-category confirmation.

The secondary result is narrower. Frozen text features improve the repaired Musical_Instruments frequency-tail endpoint (+0.000420 [0.000181, 0.000660]), but the gain concentrates at train frequency 5; removing that boundary group leaves no detected effect. The corresponding cross-dataset contrast is nonsignificant, and thinning interventions do not identify a tail mechanism. We therefore report a dataset-specific frequency-5 case, not a general cross-dataset result. TAPE, late fusion, sparse-warm redistribution, and the probe screen are supporting or descriptive studies, not additional headline contributions.

The evidence has important limits: uncertainty is over optimizer seeds on reused fixed splits; comparator uncertainty is not distributionally matched; several campaigns are outcome-visible or protocol-deviated; and no item-text permutation control, temporal split replication, or deployment study has been completed. The defensible contribution is thus modular rather than architectural: a small causal FIR realization, its measured internal effect under the stated protocol, and an auditable record that preserves nulls, deviations, and VOID outcomes alongside positive results.

8 Code and Data Availability

Code, frozen protocols, result artifacts, and build scripts are public at https://github.com/Ray0419/bestrec-sota-results. The tracked `RELEASE_MANIFEST.json` binds source code, split derivatives, text caches, result JSONs, released per-user sidecars, and the pinned HSTU-BLaIR submodule by SHA-256. Running `bootstrap_public_clone.py` fetches and verifies release-class assets; running `rebuild_hstu_submission.py --strict` then checks HSTU parity, recomputes the 190 artifact-gated cells, verifies the manifest, and executes every governed adjudicator.

The public `v0.9-audit-evidence` release is the working evidence store. Historical campaigns for which checkpoints or per-user records were not retained are labeled as such and require regeneration for rank-level reconstruction. Raw Amazon Reviews 2023 archives are not redistributed; derived splits are tied to their public source and preprocessing code by hashes. Local-only audit sidecars are outside every printed claim and are precommitted by digest for reviewer access. A new archival deposit will be cut only after the manuscript, author metadata, licenses, and clean-clone build are finalized; the older `v1.1.11-deposit` tag is retained as a historical boundary, not represented as current.

9 Acknowledgments

We thank the Amazon Reviews 2023 maintainers [Hou et al. 2024] for releasing the dataset and accompanying reference implementations. The MLP adaptor design used in our best variant is from their published SASRecText configuration.

10 Ethics and Data Governance

This work uses only the public, pseudonymized **Amazon Reviews 2023** dataset (Hou et al., 2024; McAuley Lab), obtained from its official distribution. The dataset's maintainers state publicly that they are not in a position to assign a license or dictate usage terms; **that statement is not an affirmative permission grant, and we do not treat it as one**. We redistribute derived interaction-split CSVs (never the raw dataset) with attribution on the basis of the dataset's public research availability, we will remove them immediately on maintainer, platform, or venue request, and we flag the redistribution basis for venue-level review rather than asserting a legal right. Scope of data actually consumed: interaction tuples (pseudonymous user identifiers — remapped to dense integers by our preprocessing — item identifiers, and timestamps) and **item metadata text** (titles, categories, and store/seller name — the AR2023 `store` field; an earlier label said "brand", corrected 2026-07-20: the frozen cache text template's literal `brand:` prefix is retained as-is inside the cached strings and documented as a misnomer, because regenerating the caches would invalidate every frozen-text result) for the frozen text encoders. We do **not** process review text bodies, ratings-as-text, or product images, and we make no attempt at re-identification of any user. Released artifacts follow the same boundary: per-user evaluation sidecars contain only remapped integer IDs, target item IDs, and rank outcomes; the raw dataset is **not redistributed** (our releases carry SHA256 hashes and regeneration scripts instead, `RELEASE_MANIFEST.json`). No new data was collected and no interaction with human subjects took place; no institutional review determination was sought or is claimed — the data are public, platform-pseudonymized product reviews, and we state that as a fact rather than adjudicating any jurisdiction's requirements; ACM policy places responsibility for applicable institutional, ethical, and legal compliance on authors, and we accept it, with this section's factual basis available as supporting documentation on request. **Sidecar identifier surface, stated exactly:** the released per-user sidecars carry our preprocessing's dense remapped user indices (`user_id` 0, 1, 2, ...), a target-item index, and rank outcomes — not hashes; those dense indices are deterministically linkable to the platform's pseudonymous identifiers through the released split CSVs, so the sidecars are exactly as pseudonymous as the public dataset itself, no more — we state this linkage surface plainly instead of calling the sidecars anonymous. Released derivatives are retained under the tagged GitHub release; removal requests (maintainer, platform, or a data subject acting through the platform's pseudonymization) will be honored by deleting the affected assets and regenerating the release manifest. Finally, this is an offline evaluation study: recommender systems deployed on such data can amplify popularity and exposure biases — our long-tail analyses quantify one aspect of that concern — and nothing in this paper constitutes a deployment claim.

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Appendix A.0 — Office_Products: descriptive evidence (VOID under its pre-declaration)

Second-category pre-declared confirmation: Office_Products. To test whether the per-category result generalizes, we froze a second pre-declaration (`SOTA_CONFIRM_PREREG_OFFICE.md`, committed before the raw data finished downloading — before any Office statistic was knowable) with the config carried over from Musical_Instruments **unchanged (zero category-specific tuning)** and fresh seeds 20260623–27. Dataset parity is exact (items 77,551 and users 223,308 match the comparator paper exactly; interactions differ by one, as on MI). Result, on final-epoch full-catalog evaluations (223,308 users; the prereg's declared headline rule): **K=16 = 0.03042 ± 0.00008 (95% CI lower bound 0.03032) and K=8 = 0.03033 ± 0.00018 (CI-LB 0.03010), all 10 fresh seeds above the published HSTU-BLaIR point estimate 0.0271** — a ~+12% margin, versus +2.3% on MI. **However, the pre-declared floor check FAILED: our plain-SASRec floor (0.02208) lands +44% ABOVE the published SASRec (0.0153), violating the prereg's comparability condition. Under the pre-declaration as written, the Office second-category pass is therefore VOID — the gate values are reported as a provisional, non-confirmatory result — the matched comparator-baseline run that explains the floor gap is complete and explains it (resolution paragraph below); the VOID stands. The V1 Office campaign counts in no claim.** (It leaves the pre-declared Musical_Instruments confirmation of §5.2 untouched, and the separately pre-declared, redesigned **V3** campaign — environment-matched reference, fresh never-inspected seeds — later **passed** and is counted strictly under its frozen per-category point-estimate wording, §5.2. The V1 VOID stands unchanged.) Disclosures: best-by-val checkpoints were evaluated on a 30k-user subsample (declared in the prereg), so the headline uses the always-full final-epoch eval — the conservative choice (the subsampled best-checkpoint numbers were higher); the run manifests carry a dirty-tracked flag caused by the driver's own results-file appends (embedded per-run code hashes verify protocol-code identity); our plain-SASRec floor (0.0221) lands above their published SASRec (0.0153) on this category — a baseline-config-strength difference, with split parity resting on the exact dataset-statistics identity and frozen hashes rather than the floor heuristic; and the pre-declared tail prediction for Office was scored **VOID** (its connectivity 2.89 falls in the pre-declared ambiguous zone, plus a disclosed formula inconsistency in the stats tool — `SOTA_CONFIRM_OFFICE_RESULTS.md`). Descriptively, the Office tail contrast is strongly text-positive at full catalog (pooled tail hits, text vs ID-only: 364 vs 268 @10; 1,983 vs 1,247 @100). **Statistical correction (2026-07-19):** the previously printed pooled two-proportion z statistics (z=3.8 @10, z=13.0 @100) are retracted — the same 36,610 tail users are evaluated under every seed and under both arms, so the 183,050 seed-summed rows are clustered repeated observations, not independent trials, and an unpaired Bernoulli z is not valid (nor conservative) under that dependence. Treating the five trained models per arm as the inferential units instead, independent-arm Welch tests give ΔHR@10 = +0.000524 ($t = 7.26$, df ≈ 8, $p \approx 9\times10^{-5}$, 95% CI [+0.00036, +0.00069]) and

ΔHR@100 = +0.004021 (t = 16.75, df $\approx$ 8, p $\approx 2\times10^{-7}$, 95% CI [+0.00347, +0.00457]) — same direction, valid geometry; final-evaluation per-user sidecars for a fully clustered analysis are a queued release item. This remains pattern evidence (the pre-declared tail prediction was VOID), consistent with the connectivity gradient (MI 2.34 win → Office 2.89 win → Beauty 3.51 / VG 3.70 null), not a scored prediction.

office_confirmation (regenerated): second-category pre-declared confirmation — Office_Products (SOTA_CONFIRM_PREREG_OFFICE.md; fresh seeds 20260623–27; headline rule = final-epoch FULL-catalog eval, history[-1].test, n_eval=223,308 — NOT the 30k best_test subsample).

STATUS: VOID under the prereg floor check (+44% floor inflation) — provisional; the V1 Office campaign counts in no claim. Office V3 — a separate, redesigned pre-declaration — passed and is counted under its frozen per-category point-estimate wording (§5.2).

arm	per-seed NDCG@10 (final-epoch full)	mean ± sd	95% CI-LB	vs published 0.0271
K=16	0.03041 / 0.03055 / 0.03042 / 0.03031 / 0.03043	0.03042 ± 0.00008	0.03032	ABOVE (5/5 seeds) — provisional/VOID
K=8	0.03035 / 0.03036 / 0.03048 / 0.03043 / 0.03002	0.03033 ± 0.00018	0.03010	ABOVE (5/5 seeds) — provisional/VOID
ID-only (contrast arm)	0.02845 / 0.02833 / 0.02843 / 0.02837 / 0.02840	0.02840 ± 0.00005	0.02834	(no gate; tail-contrast arm)
SASRec floor (best_test, seed 20260623)	0.02208	—	—	floor check FAILED: +44.3% vs published SASRec 0.0153 → Office pass VOID under prereg

Fresh seeds above published 0.0271: 10/10 (provisional; not a counted pass). Pooled tail hits, text (k8 arm) vs ID-only, final-epoch full eval (tail n=36,610/seed): @10: 364 vs 268 (descriptive counts; pooled z retracted 2026-07-19); @20: 586 vs 414 (descriptive counts; pooled z retracted 2026-07-19); @50: 1170 vs 739 (descriptive counts; pooled z retracted 2026-07-19); @100: 1983 vs 1247 (descriptive counts; pooled z retracted 2026-07-19) — descriptive post-hoc pattern evidence only (the pre-declared Office tail prediction was scored VOID: connectivity 2.89 in the pre-declared ambiguous zone).

Floor-anomaly resolution (2026-07-11; the matched comparator-baseline run promised above, now complete). We executed the reference implementation's own Office SASRec configuration end-to-end — its preprocessing (dataset-size assertions passing: 77,551 items / 223,308 users), its gin config, its trainer, its eval — locally under the shimmed research path of §5.6. Its final-epoch NDCG@10 is **0.0174** (best full eval 0.0177), i.e. **+13.9% above its own published row 0.0153**, crossing the published value at epoch 20 of 101 and never re-entering. The +44% floor-check failure now decomposes exactly: 0.0204 (our floor run's final-epoch full-catalog reading) / 0.0174 (their code, locally) = **+16.9% attributable to baseline-strength protocol differences** (our floor trains with full-catalog softmax vs their 512-negative sampled softmax), and 0.0174 / 0.0153 = **+13.9% attributable to the published row sitting below what its own pipeline regenerates here** (1.139 × 1.169 = 1.331 = 0.0204/0.0153; the remaining step to the prereg's best-epoch 0.02208 is checkpoint selection plus that variant's 30k-user eval subsample). **Conclusion: the elevated floor is not an artifact of our data split or evaluation** — Office simply supports SASRec well above the published row. This *explains* the floor-check failure without repairing the comparison it protects: if the published SASRec row is conservative in this environment, the published HSTU-BLaIR 0.0271 plausibly is as well, so numerically exceeding it would not establish superiority over their method on Office. **The VOID therefore stands, deliberately** — and the pre-declared floor check is vindicated: it caught exactly the comparator-conservatism failure mode it was designed to catch.

Completed HSTU-BLaIR check (2026-07-12). The resolution above inferred that the published HSTU-BLaIR 0.0271 was "plausibly" conservative as well; running their Office HSTU-BLaIR configuration end-to-end locally **tested and refuted that inference**: it lands at final-epoch NDCG@10 **0.0275** (best full eval 0.0279), i.e. **+1.6%/+2.8% above the published 0.0271** — the flagship comparator row regenerates in this environment (margins comparable to the Musical_Instruments regeneration), unlike the SASRec row (+13.9%). Descriptively, the (VOID) gate values 0.03042/0.03033 sit ≈+9% above both the published value and its local regeneration. **The VOID nonetheless stands**: the pre-declared floor check failed as written, and no post-hoc result — favorable or not — restores a voided pre-declaration. These numbers are environment-caveated single-run regenerations, reported as descriptive evidence only; the V1 campaign recorded in this appendix remains counted in no claim. (A redesigned V3 pre-declaration — new never-inspected seeds, environment-matched reference — subsequently **passed** and is reported in §5.2; the V1 VOID stands unchanged.)

Appendix A — Superseded earlier-stage supporting material (Beauty_and_Personal_Care)

> **Why this is an appendix:** the material below is an earlier-stage Beauty_and_Personal_Care reproducibility study (the 20-variant cross-pipeline scan, its 2-seed signal, and the retracted LIGER-gap audit). It predates and does not belong to the headline spine (causal FIR filter + MI frequency-5 tail case + competitive-overall Video_Games result, §1–§7). It is retained verbatim for the reproducibility and negative-result record only, and is referenced from §6.1–§6.2 as supporting context. The live Beauty result that enters the tail pattern (§5.3) is the text–ID tail contrast, not these single-seed sampled-softmax scans.

A.1 Beauty_and_Personal_Care — 20-variant cross-pipeline scan (superseded supporting material)

> **Note:** the following Beauty_and_PC 20-variant scan and the LIGER-gap audit are **earlier-stage supporting material**, retained for the reproducibility and negative-result record. They predate the HSTU + causal-filter + tail-pattern spine and are *not* the paper's headline; the live Beauty result used in the tail pattern (§5.3) is the text–ID tail contrast, not these single-seed sampled-softmax scans.

The scan varies five axes: **text encoder** (MiniLM titles/rich-text, BLaIR titles/rich-text), **projection** Φ (single Linear vs the 2-layer MLP adaptor of Hou et al. 2024), **learned item embedding** (kept vs removed = faithful SASRecText), **loss** (sampled softmax K=1024, in-batch negatives, chunked-full-softmax, hybrid), and **augmentation** (random subsequence sampling, factor 1/2/3). Training used a 15–20-epoch budget, batch size 256, Adam (lr 1e-3, weight decay 1e-5, gradient clip 5.0), a single seed for the initial scan, and two seeds (42, 43) for the final reported variants. Table A1 summarizes all 20 variants. **Numbers are single-seed unless otherwise noted.**

Table A1: Beauty_and_PC 5-core cross-pipeline scan (full-catalog eval, NDCG@10) — 22 numbered rows: 20 distinct variants plus two seed replications (row 20 replicates variant 19; row 22 replicates baseline variant 7)

#	Variant	NDCG@10	Δ vs base
1	Popularity	(not run)	—
2	SASRec sampled-1024 + MiniLM titles	0.0180	—
3	SASRec sampled-1024 + MiniLM titles, d=128	0.0176	-2.2%
4	SASRec in-batch negs + augment-2	0.0036	-80%
5	TIGER minimal (our re-impl)	0.0080	-56%
6	SASRec hybrid (in-batch + 8K random negs + aug-3)	0.0106	-41%
7	**SASRec chunked-full + MiniLM titles (improved baseline)**	**0.0191**	**+6.1%**
8	SASRec chunked-full + MiniLM titles, d=128	0.0186	-2.6% (vs 7)
9	SASRec chunked-full + MiniLM titles, 25 epochs	0.0189	-1.0% (vs 7)
10	BERT4Rec	0.0160	-16% (vs 7)
11	SBERT-only (no learned item_emb) + MiniLM	discontinued	—
12	Faithful SASRecText (no item_emb + MLP + dropout 0.5 + BLaIR)	discontinued @ epoch 2	—
13	Hybrid (item_emb + MLP + dropout 0.5 + BLaIR)	discontinued @ epoch 5	—
14	BLaIR-titles + Linear projection (just encoder swap)	discontinued @ epoch 2	—
15	Fusion (BLaIR + rich-text MiniLM concat) + Linear	discontinued @ epoch 1	—
16	**BLaIR-titles + MLP adaptor (Hou et al. [2024]) + dropout 0.2**	**0.01928**	**+0.9% (vs 7)**
17	+ augment-2	0.0036 val	discontinued; aug hurts
18	+ d=128 width	discontinued at eval (slow)	—
19	**BLaIR-rich-text + MLP adaptor + dropout 0.2 (seed 42)**	**0.01936**	**+1.3% (vs 7)**
20	seed 43 of variant 19	**0.01955**	**+2.3% (vs 7)**
21	**MiniLM + MLP adaptor (clean ablation)**	**0.01889**	**-1.2% (vs 7)**
22	Baseline seed 42 (reproduction)	0.0190	0.0% (vs 7)

2-seed mean of variant 19+20 (BLaIR-rich-text + MLP): NDCG@10 = 0.01946, range 0.0002.

Clean-ablation finding (variant 21, added after audit): When we hold the encoder constant at MiniLM and ONLY change projection layer from single-Linear to the SASRecText 2-layer MLP adaptor, NDCG@10 drops slightly from ~0.0190 to 0.01889. **The MLP adaptor by itself does NOT help on this protocol.** The +1.3-2.3% gains we observe with variants 16/19/20 are attributable to the change in text encoder (MiniLM → BLaIR) and the change in text content (titles → rich text), not to the projection layer.

Re-analysis of the 20-variant scan:

- **MLP adaptor alone**: no benefit (variant 21 vs 7 = -1.2%)
- **MiniLM → BLaIR encoder**: small benefit (variant 16 vs 7 = +0.9% NDCG@10, single-seed)
- **Titles → rich text (BLaIR)**: small additional benefit (variant 19 vs 16 = +0.4%, single-seed)
- **Removing learned item embedding** (faithful SASRecText): hurts substantially
- **Heavy dropout (0.5)** as in the published SASRecText: hurts on our 5-core data (median 5 interactions/user)
- **Subsequence augmentation**: hurts (~-5% NDCG)
- **Width (d=128)**: no help

A.2 Beauty_and_PC multi-seed signal (superseded supporting material)

> **Note:** this is appendix supporting material for the Beauty 20-variant scan (§A.1), not the headline tail-pattern result (§5.3).

We have multi-seed numbers for two configurations:

- Best variant (BLaIR + rich-text + MLP): seed 42 → 0.01936, seed 43 → 0.01955; **mean 0.01946 (n=2)**.

- Baseline (chunked-full, no MLP, MiniLM titles): original → 0.01911, seed 42 → 0.0190; **mean 0.01906 (n=2)**.

Apparent gain: 0.01946 - 0.01906 = **+0.0004 (+2.1% relative)** with n=2 vs n=2. Inter-seed variance for our baseline (0.0001 range) is roughly half the size of the gain, suggesting the directional signal is real but small. A formal significance test would require larger n; our paper-stated +2.1% improvement is honest about this.

A.3 Audit of "gaps to LIGER" claims (corrected from earlier versions; superseded supporting material)

> **Note:** this comparator-audit retraction is retained as appendix material; the body's §5.4 is the titration result.

Earlier versions of this work cited an apparent 57% gap to LIGER on Beauty_and_Personal_Care. **After a comparator audit, this framing is retracted.** The facts:

1. **LIGER reports on Amazon 2014, not AR2023.** The LIGER paper [Yang et al. 2024] evaluates on Amazon Beauty (2014, He and McAuley 2016), reporting NDCG@10 = 0.04020 ± 0.00044 (K=20) and 0.04738 ± 0.00151 (K=N) on their 5-core preprocessing. The 2014 Amazon Beauty subset has tens of thousands of items, while our AR2023 Beauty_and_Personal_Care 5-core has 207,649 items. **These are different datasets.** The "57% gap" framing in earlier versions compared our AR2023 number to LIGER's 2014 number, which is invalid.
2. **There is no published LIGER number on AR2023 Beauty_and_Personal_Care 5-core.** LIGER does not evaluate on AR2023 at all in the published paper.
3. **TIGER reports on Amazon 2014 only.** The TIGER paper [Rajput et al. 2023] reports NDCG@10 = 0.0384 on Amazon Beauty 2014 with their 5-core, also not on AR2023.
4. **BLaIR [Hou et al. 2024] introduced AR2023 — historical note, superseded: the AR2023 5-core literature has since grown (HSTU-BLaIR and 2026 preprints; §5.1).** Their Beauty subset is "All_Beauty" (not "Beauty_and_Personal_Care") and their preprocessing is by-timestamp 8:1:1 with no k-core filter. They report NDCG@10 on All_Beauty in the range 0.0177-0.0241 across various LLM encoders (Table 8). With a UniSRec downstream model, their best All_Beauty number is 0.0241 (gemini-embedding). They do not report on Beauty_and_Personal_Care.

We therefore cannot make a defensible gap claim to TIGER/LIGER/BLaIR on AR2023 Beauty_and_Personal_Care 5-core, because **we did not find a comparable published number** (a search statement as of this writing, not a priority claim — AR2023 preprints are appearing rapidly; §5.1). We provide our 0.01946 (2-seed mean) as a reference baseline rather than as a SOTA-improvement claim.

A future apples-to-apples comparison would require either:

- Running TIGER/LIGER on our 5-core preprocessing (multi-week effort for faithful reproduction), or
- Adopting Hou et al.'s by-timestamp 8:1:1 preprocessing and re-running our pipeline.

S.4 — Screening-log detail (full probe table)

> *Supplement to §5.5 — the full search record. Each row's Δ is versus its stated base; most rows are single-seed exploratory probes (power note in §5.5).*

Table S1: Screening log — capacity-adding / alternative-mechanism probes; none showed a benefit at tested power (mostly single-seed; allocation was partly outcome-dependent, so this is a search record, not a systematic sweep). (Mechanism class: where the lever acts. "Learned scalar → 0?" = did a free learnable gate/temperature/intensity collapse the mechanism. n = seed count. Probe powers are unequal — see the paragraph above.)

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar → 0?	Verdict (observational at n=1)
c3 continuous time-decay attention kernel	attention re-weighting (capacity-add)	SBERT stack ≈0.0639	**−0.0150** (1; test 0.0489 at record val 0.0725)	n/a	No benefit observed (single-seed probe) — actively harmful; the largest validation–test divergence in the corpus
Sampled softmax (K=512 negatives)	loss approximation	full-softmax stack	−0.0026 (1)	n/a	No benefit observed (single-seed probe) — undertrains full-catalog ranking
Dual text encoder (SBERT ⊕ BLaIR)	text encoder	SBERT stack	−0.0014 (1)	n/a	No benefit observed (single-seed probe) — the text encoder is not the gap
BLaIR text encoder (swap)	text encoder	SBERT stack	−0.0013 (1)	n/a	No benefit observed (single-seed probe) — encoder axis fully swept
GD1 spectral-shrink prior	representation regularizer	V2 0.0673	−0.003 (1)	shrink → 0	Rejected (the formerly linked spectral figure is retracted, §5.4 retraction note)
n_heads = 4	architecture (capacity-add)	HSTU-style stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) — ties 2-head even at the published dv=dqk=16 spec
CL4SRec self-supervision	SSL auxiliary loss	SBERT stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) — early-epoch only
c1 prototype-routed expert heads	capacity-add	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) — redundant with TAPE
c2 text-distillation aux loss	auxiliary loss	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) — redundant with the additive text path
text-init / text warm-start	embedding initialization	SBERT stack	−0.0003 (1)	n/a	No benefit observed (single-seed probe) — redundant with the additive text path
EMA / SWA weight averaging	weight-space regularizer	SBERT stack	+0.0001 (1)	n/a	No benefit observed (single-seed probe) — val-high/test-flat; does not repair the structural gap
text-sim bias	attention bias	HSTU-style stack	±0.0001 (4)	n/a	NEUTRAL — no benefit observed; removable candidate
W1 niche-share fitness-sharing penalty	inter-item loss penalty	U2/LS0.2 0.0649±0.0002	0.0652 (1), within band	$\beta = -$**7.42** (sign-flipped: boosts, not penalizes, crowded niches)	No benefit observed (single-seed probe) — refutes the anti-crowding remedy
X1 frequency-adaptive James–Stein shrinkage	representation regularizer	V2 0.0674±0.0003	0.0673 (1), within band	**c** ≈ **0** (λ_max ≈ 0.004) → off	No benefit observed (single-seed probe) — model voted shrinkage off
Y1 heat-kernel/manifold label smoothing	loss-target reshaping	V2 0.0674±0.0003	0.06653 (1), ≤ band	**T** → **0.00061** (< init) → delta target	No benefit observed (single-seed probe) — text-manifold geometry voted off
Z1 forced ID→text routing	representation routing	V2	tail −75% (globally dead)	forced gate	Rejected — globally harmful
CF1 cue-fusion gate	representation fusion	V2 / MI	flat/dead (both datasets)	gate → 0	Rejected — new-lever search closed
conn-gate connectivity-gated cold-start ID↔text fusion	representation fusion/routing	V2 text stack, MI	**tail** Δ +0.0001 ± 0.0003, same-seed 95% CI [−0.00025, +0.00045] (5); overall flat (cadence-caveated, §5.5 text)	α → **0.0011** < **init 0.0025** (voted off; recorded in 4/5 seed logs)	Rejected — operationalizes the intervention-associated connectivity tail pattern, but is not actionable (CI includes 0)

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar → 0?	Verdict (observational at n=1)
max_seq_len 200 / seq > 50	sequence length	SBERT stack	~0 (1)	n/a	No benefit observed (single-seed probe) — sequences short post-5-core
cosine scoring	scoring function	winning op-point	≤ 0 (1)	n/a	No benefit observed (single-seed probe) — dot-product better at the op-point

S.5 — Titration ladder (full six rungs)

Supplement to §5.4; integrity-gated against the §5.4 prose values by `_bestrec_run/make_table_5_4_titration.py`.

Table S2: the interaction-thinning density-titration ladder (AR2023 Video_Games 5-core LLOO, full-catalog n_eval = 94,762, tail_n = 10,900; same-seed-number text–ID (not initialization-paired, §5.3), best-by-val; 5 seeds = 20260608–12 per rung; mean ± sample-std, positive-seed count in parens). Realized interactions/item are the run-log values. (The former α = ipi/d_eff column is retracted with the spectral analysis — see the §5.4 retraction note; its cells are removed from the paper and marked REMOVED_FROM_PAPER in the artifact graph.) All cells are re-read from the frozen `results_TITR*/TAIL_*` JSONs by `_bestrec_run/make_table_5_4_titration.py`, which **integrity-gates** the recomputed NDCG head/tail means against the locked prose values above (aborts on any >5e-5 drift); the HR columns and per-rung seed bands are the net-new tabulation the prose summarized only as Spearman trends.

ρ	kept inter./item	head ΔNDCG@10	head ΔHR@10	tail ΔNDCG@10	tail ΔHR@10
1.00	24.405	+0.002213 ± 0.000241 (5/5)	+0.004238 ± 0.000765 (5/5)	−0.000148 ± 0.000179 (2/5)	+0.000128 ± 0.000582 (3/5)
0.94	22.947	+0.002389 ± 0.000267 (5/5)	+0.004083 ± 0.000830 (5/5)	+0.000165 ± 0.000370 (4/5)	+0.000807 ± 0.000712 (5/5)
0.91	22.210	+0.002544 ± 0.000491 (5/5)	+0.004940 ± 0.000892 (5/5)	+0.000039 ± 0.000310 (3/5)	+0.000110 ± 0.000766 (3/5)
0.88	21.484	+0.002520 ± 0.000421 (5/5)	+0.004869 ± 0.000765 (5/5)	+0.000537 ± 0.000386 (4/5)	+0.000661 ± 0.000995 (4/5)
0.78	19.030	+0.002664 ± 0.000398 (5/5)	+0.004467 ± 0.001082 (5/5)	+0.000056 ± 0.000552 (2/5)	+0.000239 ± 0.001087 (2/5)
0.66	16.109	+0.003540 ± 0.000416 (5/5)	+0.005720 ± 0.000904 (5/5)	−0.000108 ± 0.000503 (1/5)	+0.000073 ± 0.001165 (3/5)